Remember What You Thought, Not What You Said: Workspace-Gated Episodic Memory for LLMs

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"We can know more than we can tell."
— Michael Polanyi, *The Tacit Dimension* (1966)

Abstract

Memory-limited LLM agents decide what to store by asking the model what is important, the self-rated importance scores introduced by Generative Agents and inherited by most agent memory systems. Under experimental control of content provenance, we show this practice fails exactly where memory matters most: information the model inferred while reading but never wrote down. On sparsely cued construct-valid benchmarks, verbal importance ratings fall at or below chance as a ranking signal, while the same models demonstrably carry the inferred content: a training-free workspace readout of the residual stream, derived from the J-lens construction, recovers it on one-hop designs (AUC 0.63–0.67 at 7B, three model families), and causally patching the corresponding directions flips memory-based answers where sham directions do not (0.70 vs. 0.05 at 7B, p=2.4×10^{-4}). The model knows more than it tells, so we gate memory on what it knows: a training-free write gate that scores candidates with the workspace readout instead of an importance probe costs zero probe passes, matches the canonical 1–10 rating downstream while remaining predictive on construct-valid content where the rating falls to chance, shows no detectable difference from the selection oracle at 7B on the construct-valid benchmark (0.525 vs. 0.525; 95% CI on the paired difference $[-0.09, +0.08]$), and significantly beats the binary probe variant it is dropped into (p=0.0013 there; p=0.0004 at 0.5B on the mixed stream). The telling is also repairable: $\sim$500 self-distillation steps on the model's own workspace rankings lift held-out verbal calibration from $\approx$0.3 to 0.62–0.65 on average across seeds (best 0.75) in both families tested. Agents should remember what they thought, not what they say they thought.

1 Introduction

An LLM agent that operates over days cannot keep everything in context. Since Generative Agents (Park et al., 2023), the standard answer to the question of what to store has been to ask the model itself: rate the importance of each candidate memory and keep the top-rated ones. Variants of this reflection-based admission control run inside most deployed agent memory stacks (Packer et al., 2023; Zhong et al., 2023). The practice rests on an assumption that has never been tested under experimental control of content provenance: that a model's verbal report of what is important tracks what will actually be useful.

We test this assumption under experimental control of content provenance and find that it fails for the content that memory selection most needs to capture. We construct episodic benchmarks in which the single most useful item is either (i) stated explicitly in the text, or (ii) a silent inference: a bridge concept ("the city is Madrid") that the model can be shown to carry while reading (§5.4) but that never appears as a word and is required only later, when a probe question arrives (in the construct-valid designs the bridge is also never the answer itself, §3). Asked which concepts are worth remembering, models at every scale we tested (0.5B–7B) and at every post-training stage (SFT, DPO, RLVR) confidently nominate vivid, useless surface words and reject the silent bridge: on the sparse-cue construct-valid benchmarks, verbal importance is at or below chance as a ranking signal (AUC 0.17–0.49; Table 1; Fig. 1). The same reports are good when the useful content is explicit, better there than any internal signal we test. The failure is robust to probe wording, to the canonical 1–10 rating form, and to letting the model deliberate before answering; its one measured boundary is cue density, summarized in §5.1 and mapped in Appendices J and I.

The model knows more than it tells: the information is present in its internal state but not in its report. Building on the J-lens "workspace" construction of Anthropic Interpretability Team (2026), we read each concept's availability

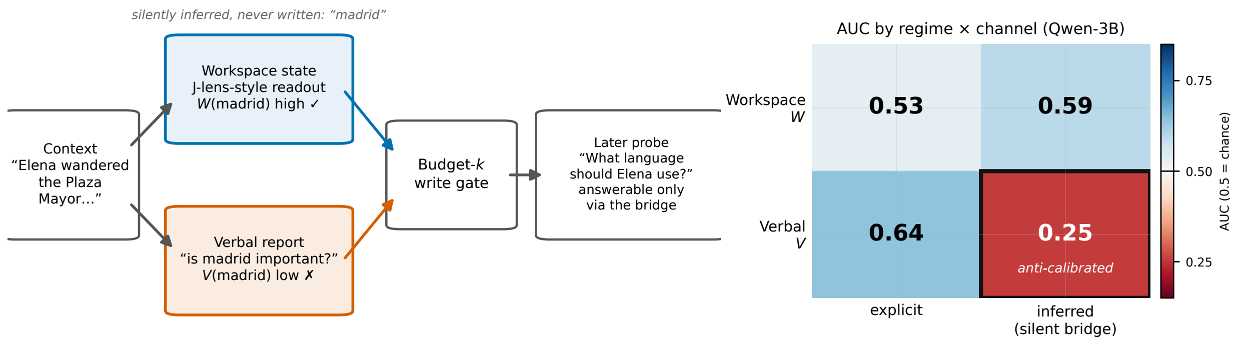

Figure 1: Left: the write-gating problem. Reading "Elena wandered the Plaza Mayor…", the model silently infers *madrid*, and the inference is visible in a J-lens-style workspace readout of its residual stream; asked whether *madrid* is worth remembering, it confidently says no. A budget-k memory gate that trusts the verbal report discards exactly the item needed to answer "What language should Elena use?". **Right:** AUC of each channel at identifying truly load-bearing items (Qwen-3B shown; same qualitative dissociation at 3B–7B in all three families, with knife-edge small-scale cells noted in App. O; Table 1): the verbal report is good for explicit content but below chance for construct-valid silently inferred content (boxed cell; generator analysis in §5.1); the workspace readout is not.

directly off the residual stream at the end of encoding: the peak reciprocal rank of the concept's token across layers under the logit lens (nostalgebraist, 2020). This readout ranks the silently inferred bridges that self-report rejects above the competing candidates (AUC 0.63–0.67 at 7B across three model families), is not explained by surface presence, embedding relevance, or prompt wording (App. J), and is causally load-bearing: steering the bridge's input-side direction while the model answers from memory changes the answer far more often than a sham direction (0.70 vs. 0.05 flip rate at 7B at the gentlest tested scale, 0.60 vs. 0.10 at 3B; Table 2).

Downstream, gating memory on the workspace score costs zero probe passes, matches the canonical 1–10 rating everywhere the two can be compared, and matches the selection oracle at 7B on the construct-valid benchmark (0.525 vs. 0.525; p=0.0013 against the binary probe; Fig. 4); against that probe it also raises 0.5B recall-QA from 0.45 to 0.75 on Evoked ($p<10^{-4}$). Full comparisons are in §5.5. Finally, the failure is repairable: the reporter never learned to read the workspace, and 500 steps of fine-tuning on labels derived from the model's own workspace ranking, with no human annotation, lifts held-out verbal calibration from ≈0.3 to 0.62–0.65 AUC on average across seeds while leaving the workspace channel and QA accuracy statistically unchanged (Table 4).

Contributions. (1) A content-provenance regime map for LLM introspective salience: verbal importance reports are reliable for explicit content and fall at or below chance for construct-valid silently inferred content, replicated across three model families and three probe wordings, with episode-cluster bootstrap statistics and a machine analogue of metacognitive efficiency (§3, §5). We release the four provenance-controlled benchmarks, show that existing naturalistic benchmarks structurally cannot exercise the silently inferred regime (App. N), and map the regime's boundary with a measured cue-density analysis across three generators (App. I). **(2)** A training-free workspace write gate, the first use of a J-lens-style readout as a write-admission signal, with causal evidence that memory-based answers read the patched workspace content and budgeted-recall payoff up to selection-oracle level at zero probe cost; a provenance-routed extension (PRG) covers mixed streams at tight budgets (§4, §5.5). **(3)** A post-training diagnosis: in our stage sweep, none of 11 transitions (8 Qwen base→instruct pairs; OLMo-2 SFT→DPO→RLVR) improves the workspace channel, and the verbal channel's miscalibration, already present in raw-probed base models, persists or worsens; we propose and validate metacognitive alignment, a self-distillation procedure that repairs the verbal channel, in two model families (§6). **(4)** A boundary analysis showing what the static workspace does and does not hold: a trained future-token lens cannot rescue compositional (two-hop) bridges, but a utility-gradient readout, the infinitesimal version of our causal intervention, can in the Qwen family (paired +0.125 over the static readout at 7B; family-dependent, §6, App. F).

2 Related Work

Agent memory and admission control. Reflection-based importance scoring originates with Generative Agents (Park et al., 2023) and persists through MemGPT (Packer et al., 2023), Mem0 (Chhikara et al., 2025), and agentic-memory successors (Xu et al., 2025); retrieval-centric systems such as HippoRAG (Gutiérrez et al., 2024) sidestep write admission rather than solve it. Internal signals have gated segmentation (surprisal event boundaries in EM-LLM, Fountas et al., 2025), token retention (entropy in SirLLM, Yao et al., 2024), and parametric test-time memory (gradient surprise in Titans, Behrouz et al., 2025), and latent prediction error gates robot episodic memory (Gorlo et al., 2026); none score concept-level utility from the residual stream, and none compare against the model's own report. Concurrent admission-control work replaces the verbal 1–10 rating with surface features (Zhang et al., 2026; Chen and Cheng, 2026) or with retrieval-time behavioral interventions (Srivastava, 2026), and circuit analysis has been used to diagnose memory operations (Mao et al., 2026); we instead gate writes from a mechanistic readout at encoding time and explain why the verbal rating fails (below-chance ranking of construct-valid inferred content), outperforming the tested surface-feature fusion proxy (App. K). These systems differ in how memories are stored, organized, and retrieved, but agree on what a memory is: a span of text the agent read or emitted, admitted by asking the model about its importance. Our question is prior to theirs: what should count as the memory of an episode at all. Human episodic memory does not archive transcripts; it stores what the reader understood, including inferences never present in the text (Bartlett, 1932; Reyna and Brainerd, 1995). The provenance-controlled benchmarks make that distinction operational for LLM agents, and the central finding is that the content most specific to comprehension, the silently inferred bridge, is exactly what verbal importance scoring discards on the construct-valid benchmarks. Reading the admission signal off the computation itself rather than off the emitted surface is therefore a claim about what agent memory should be made of, not an implementation detail; the gate is one module-level consequence that any of the systems above can adopt.

Reading internal states. Our readout is built from the logit lens (nostalgebraist, 2020) and the J-lens workspace construction of Anthropic Interpretability Team (2026), which established that verbalizable intermediate concepts occupy a decodable residual workspace and can be causally swapped; tuned and future-token lenses (Belrose et al., 2023; Pal et al., 2023), patchscopes (Ghandeharioun et al., 2024), and representation engineering (Zou et al., 2023) provide the underlying techniques; latent multi-hop reasoning over bridge entities is known to be partial and layer-dependent (Yang et al., 2024; Biran et al., 2024), which motivates measuring availability instead of assuming it. We claim no novelty for the construct; our contribution is its application to write admission, the regime-controlled comparison against self-report, and one finding in tension with the "verbalizable" framing: globally decodable content can remain unreportable (§5), the machine analogue of availability without access.

Introspection and self-knowledge. Models partially know what they know at the answer level (Kadavath et al., 2022; Azaria and Mitchell, 2023) and can be trained to report on themselves (Binder et al., 2024); behavioral work shows LLM salience is consistent yet not introspectable (Trienes et al., 2025), chain-of-thought explanations are unfaithful (Turpin et al., 2023), and internal states encode more than outputs reveal for truthfulness (Orgad et al., 2024); injected concepts can be detected internally while being verbally denied (Lindsey, 2025; Pearson-Vogel et al., 2026), and hidden states outperform self-assessment for tool-call gating (Wu et al., 2026). We extend this family to memory utility with a sharper failure mode (anti-calibration, not noise), a provenance-based regime map explaining when self-report works, causal grounding, and task payoff. The formal framing borrows metacognitive sensitivity from psychophysics (Fleming and Lau, 2014; Nelson and Narens, 1990); the cognitive parallels (judgments-of-learning driven by surface cues, Koriat, 1997; blindsight, Weiskrantz, 1986; global-workspace access, Dehaene et al., 2011) are developed in App. Q.

3 Benchmarks and Measurement Protocol

3.1 Problem Setting

Let $\mathcal{M}$ be a frozen language model and let an *episode* be a tuple $e = (x, C, q, a)$: a context passage x, a candidate set $C = \{c_1, \ldots, c_m\}$ of concepts that could be written to memory, and a probe question q with gold answer a that arrives only after x has left the context window. A *write-admission policy* with budget k is a map

$$\pi_k : (x, C) \mapsto K \subseteq C, \qquad |K| \leq k, \tag{1}$$

constrained to observe only the encoding-time state, with access to neither q nor a. At recall, the model answers from the kept set alone, and the policy's value is its downstream accuracy,

$$\mathrm{Acc}_k(\pi) \;=\; \mathbb{E}_e\big[\,\mathbf{1}\{\mathcal{M}(q_e \mid \pi_k(x_e, C_e)) \simeq a_e\}\,\big], \tag{2}$$

where $\simeq$ denotes answer match under the grader of App. M. Each candidate carries a latent utility label $y(c) \in \{0, 1\}$ (y=1 iff c is required to answer q; we call such items *load-bearing*). A policy reduces to a scoring function $S(c; x) \in \mathbb{R}$

ranked within episodes, and its selection quality is the probability of ranking a load-bearing item above a non-load-bearing one,

$$\mathrm{AUC}(S) \;=\; \Pr\big[\, S(c^+; x^+) > S(c^-; x^-)\,\big] \;+\; \tfrac{1}{2}\,\Pr\big[\, S(c^+; x^+) = S(c^-; x^-)\,\big], \tag{3}$$

where (c^+, x^+) and (c^-, x^-) are drawn uniformly from the pooled item populations with $y{=}1$ and $y{=}0$ respectively, and exact ties (common for the rank-valued scores below) receive half credit.

Provenance. The variable this paper manipulates is the *provenance* of the load-bearing concept,

$$\rho(c, x) \;=\; \mathbf{1}\{c \text{ occurs as a surface form in } x\}, \tag{4}$$

distinguishing content that is *stated* ($\rho{=}1$) from content that is *silently inferred* ($\rho{=}0$): derivable from x but never written. Our central question is how the reliability of different scoring channels depends on ρ.

3.2 Benchmarks

Why controlled construction is necessary. Answering the question above imposes two requirements on a benchmark: the provenance of the load-bearing item must vary across episodes, so that episodes exist whose outcome hinges on a $\rho{=}0$ concept, and which candidate is load-bearing must be known independently of every channel under test. Existing naturalistic memory benchmarks satisfy neither, and not by accident but by construction: their candidate memories are surface content. In LongMemEval (Wu et al., 2025) the natural memory units are the conversation turns themselves, verbatim substrings of the context, so every candidate has $\rho{=}1$; in LoCoMo (Maharana et al., 2024) the annotated observations summarize stated dialogue content (a median of 83% of a note's content words appear verbatim in the source session). On such streams the regime in which the channels could dissociate never occurs, and empirically no question-blind admission policy separates from random selection on either benchmark, while an oracle given the evidence location doubles their accuracy (App. N). Measuring the effect of ρ therefore requires constructing the variation deliberately, which is what our benchmarks do.

Each episode contains one load-bearing concept, two to three vivid but useless DISTRACTOR concepts present in x, and neutral FILLER items. The four benchmarks differ only in the provenance and role of the load-bearing item: **Explicit** ($\rho{=}1$; stated in text; 88 episodes / 417 items); **Evoked** ($\rho{=}0$; a one-hop silent bridge, e.g. x mentions Oslo and c^+ is *norway*; 75/352); **Decoupled** ($\rho{=}0$; as Evoked, but the gold answer a is one further knowledge hop from c^+, with $a \notin C$, $a \notin x$, and q sharing no content word with x except a person's name, so neither the probe nor any stored surface item reveals a; 68/335, plus a 120-episode scaled version, **Decoupled-L**); and **Compositional** ($\rho{=}0$; the bridge is only weakly evoked and matters solely through a two-hop composition; 52/261; its probes carry surface anchors, so it is construct-valid for encoding-time prediction only, App. A). We call Decoupled and Compositional *construct-valid*: by construction, neither the probe nor any stored surface item reveals the answer, so recall success must pass through the inferred bridge. Evoked is *answer-coupled*: its bridge is also the gold answer, which makes it the most sensitive design but lets stored surface content occasionally substitute for the bridge. Benchmarks are LLM-generated under these constraints and mechanically re-validated (bridge absent from x; Decoupled: a absent from x and C; probe anchor-free), then concept-deduplicated; construction details and a leakage audit appear in App. A. As a sanity check, the surface predictor $S{=}\rho$ carries no signal on Explicit (AUC=0.48), and dropping every audit-flagged near-miss episode changes no conclusion. On the inferred-regime benchmarks the planted bridge is by design nearly the only context-absent candidate, so bare absence ($1{-}\rho$) trivially separates it at the episode level; the workspace signal is not bare absence: among purely absent candidates the readout ranks the evoked bridge far above matched absent foils, and the confound analyses of App. J separate evocation from absence and from semantic relatedness.

3.3 Two Channels for Scoring Utility

Workspace channel. The J-lens construction (Anthropic Interpretability Team, 2026) established that verbalizable intermediate concepts occupy a decodable residual-stream workspace: reading intermediate layers through the model's own unembedding surfaces concepts the model is carrying, and swapping the corresponding directions swaps downstream behavior. We inherit exactly that measurement and contribute its use as a write-admission signal. Let $h_\ell(x) \in \mathbb{R}^d$ be the residual-stream state at the final position of x in layer ℓ, and let $z_\ell(x) = W_U\,\mathrm{LN}(h_\ell(x)) \in \mathbb{R}^{|\mathcal{V}|}$ be its logit-lens decoding through the model's own final norm and unembedding (nostalgebraist, 2020). Writing $\mathrm{rank}_\ell(t)$ for the rank of token t in $z_\ell(x)$, the workspace salience of a concept c is its peak reciprocal rank,

$$W(c; x) \;=\; \max_{\ell \le L}\; \max_{t \in T(c)}\; \frac{1}{\mathrm{rank}_\ell(t)}, \tag{5}$$

where $T(c)$ is the small set of case-variant first tokens of c (App. M). W is training-free, requires no forward passes beyond encoding, and is read blind to q; reciprocal rank rather than probability keeps the score scale-free across a $\sim$150k-token vocabulary.

Verbal channel. The same model is asked directly. With yes/no token sets Y, N and the importance probe $u(c, x)$ of App. J,

$$V(c; x) \;=\; \frac{\sum_{t \in Y} p_{\mathcal{M}}(t \mid u(c, x))}{\sum_{t \in Y \cup N} p_{\mathcal{M}}(t \mid u(c, x))}, \tag{6}$$

a binary-probability proxy for the Generative-Agents-style importance score; the canonical 1–10 rating variant is evaluated in App. J and is likewise at or below chance on the construct-valid benchmarks. Base checkpoints are additionally probed with a template-free variant V_{raw}. Baselines include embedding relevance $E(c; x) = \cos(\phi(c), \phi(x))$ under a sentence encoder ϕ (bge-small-en-v1.5, CLS pooling), recency, frequency, a surface-feature fusion proxy (App. K), and three paraphrases of the probe u.

3.4 Evaluation Protocol

The primary predictive metric is Eq. (3) per channel, benchmark, and model, with item-level bootstrap CIs in tables and episode-cluster bootstrap CIs (B=4,000, resampling episodes) for all headline contrasts; the four pre-specified primary contrasts ($W - V$ on Decoupled at each size) are Bonferroni-corrected. By loose analogy with metacognitive-sensitivity analysis (Fleming and Lau, 2014), each cell is summarized by a descriptive ratio

$$M_s \;=\; \frac{\mathrm{AUC}(V) - 0.5}{\mathrm{AUC}(W) - 0.5}, \tag{7}$$

reported only where $\mathrm{AUC}(W) \geq 0.55$; $M_s < 0$ indicates that the verbal report ranks truly load-bearing items below the rest even though the signal is present in the system (M_s is negative throughout the construct-valid regime; on answer-coupled Evoked it can be positive). Throughout, "below chance" refers to ranking AUC, not probabilistic calibration in the forecasting sense. Downstream, Eq. (2) is estimated per policy and budget with per-episode correctness logged for every condition; policies are compared on identical episodes with exact McNemar tests, bounded by no-memory and full-context references. Models: Qwen2.5 0.5B–7B (Qwen Team, 2025), OLMo-2 1B/7B with released SFT/DPO/RLVR stages (OLMo Team, 2025), Mistral-7B (Jiang et al., 2023); greedy decoding; a single fixed harness for all checkpoints so that lens bias cancels in contrasts.

4 Gating Memory on the Workspace

We now turn the diagnosis into a deployable mechanism: gate memory on the workspace readout. The core form is a drop-in replacement, substituting $F_W(W(c; x))$ for the verbal importance score in any pipeline that proposes and ranks candidates; it requires no training, no additional model, and no forward passes beyond the encoding pass the agent already performed, and it is the form behind every headline result in §5. For mixed streams at tight budgets we extend it to **PRG** (Provenance-Routed Gating), which routes each candidate to the channel that is reliable for its provenance. The design follows from three empirical facts established in §5, so each component carries an experimental justification rather than a heuristic one.

Design principles. *(P1) Provenance determines channel reliability.* Verbal importance ratings and embedding relevance are the strongest signals for surface-present content but are unreliable for silently inferred content, at or below chance on the construct-valid benchmarks of the primary generator, while the workspace readout is the most reliable single channel for that regime (Table 1; generator boundary in §5.1). The gate therefore scores by the workspace readout by default and may route surface content to a relevance channel when budgets are tight. *(P2) The routing variable is free.* Provenance $\rho(c, x)$ (Eq. 4) is computable exactly, at write time, by a substring test; the router adds no model calls and cannot itself be miscalibrated. The substring test is the exact instantiation for single-word candidates; for paraphrased or multi-word candidates any provenance estimator (lemmatized or embedding-based matching) substitutes into Eq. 8 unchanged. *(P3) Channels are calibrated by rank, not by moments.* Raw channel scores live on incomparable scales, and even z-normalization fails because the two distributions differ in shape (reciprocal ranks are heavy-tailed near zero; cosine similarities are approximately Gaussian): under z-scoring, silently inferred items systematically lose admission slots to surface items, which we verify empirically in App. P. Converting each channel to its population percentile rank over a calibration stream gives both routes identical uniform marginals, making a single threshold meaningful across routes.

Architecture figure (figures/architecture.pdf).

Panels: context encoding $\rightarrow$ candidate set; provenance test ρ; workspace route (ρ=0, Eq. 5) and relevance route (ρ=1); percentile-rank calibration; budget-k admission; memory store $\rightarrow$ recall QA.

Figure 2: The PRG write gate. Candidates proposed by the host memory system are split by the provenance test ρ (here a substring check against the source context): surface-present candidates are scored by embedding relevance, and absent (inferred) candidates by the workspace readout, the channel that tracks their utility most reliably in our experiments. Rank calibration places both routes on a common scale before budget-k admission. The gate adds no training and no model calls beyond the encoding pass.

Algorithm 1 PRG: provenance-routed write admission

Require: context x, candidates C, budget k, calibration CDFs F_W, F_E

$\{h_\ell(x)\}_{\ell \le L} \leftarrow$ residual states from the encoding pass of x
for $c \in C$ **do**
  if $\rho(c, x) = 0$ **then** ▷ inferred: not a surface form of x
    $S(c) \leftarrow F_W\big(\max_{\ell, t \in T(c)} 1/\mathrm{rank}_\ell(t)\big)$ ▷ workspace route, Eq. 5
  else
    $S(c) \leftarrow F_E\big(\cos(\phi(c), \phi(x))\big)$ ▷ surface route
  end if
end for
return top-k of C by S

The gate. Given a candidate set C for context x and budget k (Fig. 2), PRG computes

$$S_{\mathrm{PRG}}(c; x) \;=\; \begin{cases} F_W\big(W(c;x)\big) & \text{if } \rho(c,x) = 0 \quad \text{(inferred route)}, \\ F_E\big(E(c;x)\big) & \text{if } \rho(c,x) = 1 \quad \text{(surface route)}, \end{cases} \tag{8}$$

where F_W is the empirical CDF of workspace scores over the ρ=0 candidates of a calibration stream, F_E the empirical CDF of embedding scores over its ρ=1 candidates (in deployment, a running window of past writes), and the gate admits the top-k candidates by S_{PRG}. Calibrating each channel within its own route gives both routes uniform marginals on $[0, 1]$. The calibration demand is small: selection quality is unchanged in expectation with as few as $\sim$10 scored candidates per route (only the variance shrinks as the window grows), a disjoint calibration split performs as well as calibrating on the evaluation pool itself, and CDFs fitted on one pair of benchmarks transfer to a third (App. P). Algorithm 1 summarizes the general routed form; the recommended single-channel default sets both routes to the workspace score. The per-candidate cost is one substring test plus either a dictionary lookup into the already-computed logit-lens decodings (inferred route) or one sentence-encoder call (surface route); the workspace decodings themselves reuse the hidden states of the encoding pass the agent already ran. Memory is not a bottleneck either: the readout needs only the final-position residual at each layer, $O(Ld)$ per readout position independent of context length (under 0.2 MB at 7B in bf16, computable layer-by-layer during the forward pass), orders of magnitude below the KV cache the same pass maintains.

What is new. To our knowledge this is the first write gate to score memory candidates' concept-level utility from a residual-stream decodability readout (prior internal-signal gates operate on token- or event-level surprisal and entropy, §2), and PRG is the first to route admission decisions by content provenance. Both choices are grounded rather than heuristic: the routing rule implements the provenance dissociation of §5.1 (each channel is used precisely where it is empirically reliable), and the use of the workspace channel on the inferred route is supported causally: the representations it reads are the ones that memory-based answers consume (§5.4). The gate composes with, rather than replaces, existing memory systems: any pipeline that proposes candidates and ranks them by a verbal importance

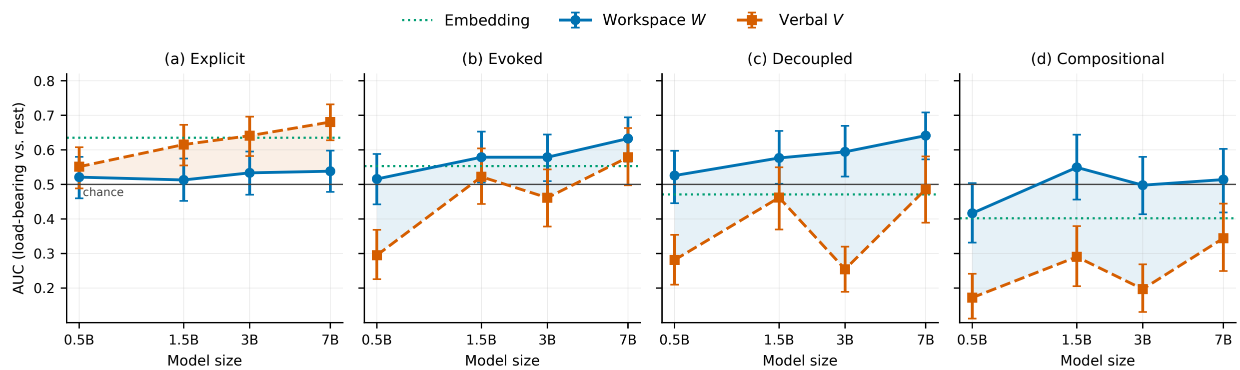

Figure 3: AUC of the workspace readout (W), the verbal report (V), and embedding relevance across model sizes (Qwen2.5-Instruct) and provenance regimes; error bars are 95% item-bootstrap CIs, the gray line marks chance. The two channels trade places as provenance shifts from explicit to inferred; on the construct-valid Decoupled benchmark the verbal report stays at or below chance at every size.

score can substitute S_{PRG} for that score unchanged. The contribution is scoped to the scoring stage; whether the host's proposer surfaces the right candidates is a separate problem, and the generative use of the readout as a proposer is beyond the plain logit lens (App. P).

Two extensions. First, the single-channel core is the recommended default: it is the strongest zero-probe-cost policy on mixed streams at capable scales, statistically tied there with the canonical 1–10 rating, which costs one probe pass per candidate; routing adds value at tight budgets and small scales, where the channels are complementary (§5.5). Second, for systems that must retain a verbal scoring interface, the metacognitive alignment procedure of §6.1 distills the workspace signal into the verbal channel with $\sim$500 self-supervised steps, repairing the deployed scorer itself rather than bypassing it. We also evaluate a third, generative extension, which harvests candidates the writer never proposed directly from the workspace (App. P), and find it beyond the reach of the plain logit lens, delineating the boundary between discriminative and generative use of the readout.

5 Results

5.1 A Provenance-Dependent Dissociation

Table 1 and Fig. 3 give the central result. On explicit content, the verbal report is the best available signal (V up to 0.681 at 7B; V exceeds W in all three families, significantly in two under cluster CIs, and embedding relevance also beats W there): when importance is written on the surface, asking the model works, and cheap heuristics work well too. The asymmetry is mechanically natural: by the end of encoding, surface mentions are largely no longer decodable at the readout position whether useful or not (median $W \approx 0.001$ for both classes on Explicit), so the readout carries little utility signal there, while the probe re-reads the full context and can evaluate exactly the surface content it sees. On evoked silent bridges the ordering reverses. On the construct-valid benchmarks (Decoupled here, Compositional in §5.2) V drops to 0.25–0.49 on Decoupled (0.17 on Compositional), at or below chance, with confident yes-ratings concentrated on vivid distractors, while W reads 0.53–0.67 on Decoupled (at chance on Compositional for two of three families, §5.2); on the answer-coupled Evoked benchmark V is more variable (above chance at 7B) yet still loses to W where the contrast is significant (Table 1). The paired difference is significant at three of four sizes on the original construct-valid Decoupled set and at 1.5B on the 120-episode scaled set built for that knife-edge case (Bonferroni-corrected cluster CIs, both intervals reported in App. O), in all three model families, and for all three probe wordings (App. J). Across three benchmark generators the two channels differ in kind: the workspace readout stays within a narrow band (0.53–0.66 on the Decoupled design) while the verbal channel swings from anti-calibrated to well-calibrated with generator and scale (0.26–0.88). The boundary variable is measurable: annotated cue density (independently evoking context elements per episode) predicts 7B reporter success (Spearman ρ=0.25, p=4 $\times$ 10^{-4} within the primary generator) while the workspace readout is flat in it (ρ=0.10, p=0.14; full analysis in App. I). The model knows more than it tells, and the telling, not the knowing, is the unstable channel: a gate can bet on the readout. In M_s terms: in the explicit regime the verbal channel dominates the workspace channel outright (V>W significantly),

Table 1: Workspace and verbal channels by provenance regime (7B, three model families). AUC of the workspace readout (W) and the verbal importance report (V) at recovering oracle-labeled load-bearing items, with the paired episode-cluster bootstrap difference (2,000 resamples; *: 95% CI excludes 0) and machine metacognitive efficiency $M_s = \frac{\mathrm{AUC}(V)-0.5}{\mathrm{AUC}(W)-0.5}$ (reported only when $W \geq 0.55$; the denominator is unstable otherwise). Verbal report wins on explicit content, is anti-calibrated on the construct-valid inferred regimes (Decoupled and Compositional, $M_s < 0$ in four of four reportable cells), and never catches up on these sparse-cue benchmarks (generator dependence: App. I).

Regime	Family	W	V	$W - V$ [95% CI]	M_s
Explicit	Qwen2.5-7B	0.538	0.681	-0.142 [-0.228, -0.048]*	n/a
Explicit	OLMo-2-7B	0.492	0.622	-0.129 [-0.211, -0.045]*	n/a
Explicit	Mistral-7B	0.532	0.622	-0.089 [-0.184, +0.003]	n/a
Evoked	Qwen2.5-7B	0.633	0.579	+0.054 [-0.034, +0.140]	0.59
Evoked	OLMo-2-7B	0.669	0.539	+0.130 [+0.049, +0.205]*	0.23
Evoked	Mistral-7B	0.670	0.589	+0.080 [-0.004, +0.167]	0.53
Decoupled	Qwen2.5-7B	0.641	0.486	+0.155 [+0.050, +0.259]*	-0.10
Decoupled	OLMo-2-7B	0.665	0.352	+0.313 [+0.229, +0.401]*	-0.89
Decoupled	Mistral-7B	0.672	0.481	+0.191 [+0.092, +0.294]*	-0.11
Compositional	Qwen2.5-7B	0.514	0.344	+0.170 [+0.071, +0.273]*	n/a
Compositional	OLMo-2-7B	0.534	0.296	+0.238 [+0.136, +0.335]*	n/a
Compositional	Mistral-7B	0.596	0.322	+0.274 [+0.186, +0.366]*	-1.85

while on the construct-valid benchmarks M_s goes negative at every scale where it is defined: the report ranks the truly useful items below the rest. Embedding relevance also fails on inferred bridges (chance on Evoked/Decoupled, 0.40 on Compositional) and the workspace beats it decisively on Decoupled ($+0.170$ $[+0.072, +0.275]$): neither self-report nor retrieval-style relevance sees what the model inferred.

5.2 Limits of Static Readouts

The workspace readout does not recover every class of inferred content, and the boundary is informative. On Compositional, where the bridge is weakly evoked and matters only through a two-hop composition, W sits at chance for Qwen and OLMo (Mistral: 0.60). This limitation is not an artifact of an untrained readout. Taking recovery to mean a paired improvement over the static readout at 7B whose 95% CI excludes zero, a tuned future-token lens initialized at the model's own unembedding and trained on disjoint contexts (Pal et al., 2023; Belrose et al., 2023) does not recover Compositional ($+0.049$ $[-0.053, +0.149]$ at 7B; its only paired gain is at 0.5B, where the absolute AUC remains near chance) and is significantly worse than the plain logit lens on evoked bridges at 7B (-0.088; App. E). Across every static readout we tested, including a conditional-likelihood score that is predictively comparable elsewhere (LL$-W$ on Compositional $+0.078$ $[-0.040, +0.188]$, not a recovery under the paired criterion; App. J), the end-of-encoding state holds inferences that the narrative actively evokes, rather than everything a future question might make relevant. §6 shows a gradient-based readout crosses this boundary in one of the two families tested.

Post-training analysis. In our stage sweep, none of 11 transitions improves the workspace channel: eight Qwen base→instruct pairs are flat to slightly negative, and OLMo-2's actual pipeline declines cumulatively (base 0.631 → SFT 0.599 → DPO 0.576 → RLVR 0.580; -0.052, directional: one-sided p=0.041, two-sided cluster CI $[-0.105, +0.007]$; App. D). What is unambiguous is the absence of improvement in any of the 11 transitions. The verbal channel's inferential-regime failure is not created by chat tuning: raw-probed base models are erratic on inferred content (e.g., Qwen-7B base V_{raw}=0.11 on Compositional; OLMo-2-1B base 0.445), and the chat-tuned probe settles at a confidently miscalibrated 0.27–0.30. Post-training rewards are computed on emitted text, never on internal availability, and the result is a fluent report channel that was never trained against the internal one. (Scope: this concerns utility-salience; Pearson-Vogel et al. (2026) show that detection of injected concepts can be elicited in models, a different construct, attributed there to in-context information rather than a post-training mechanism.)

5.3 Budget-Constrained Memory Selection

Fig. 4 evaluates the two channels as write-admission policies. At 0.5B on evoked bridges, gating on the binary importance probe stores confidently wrong vivid words and at k=1 performs at the no-memory floor, while workspace

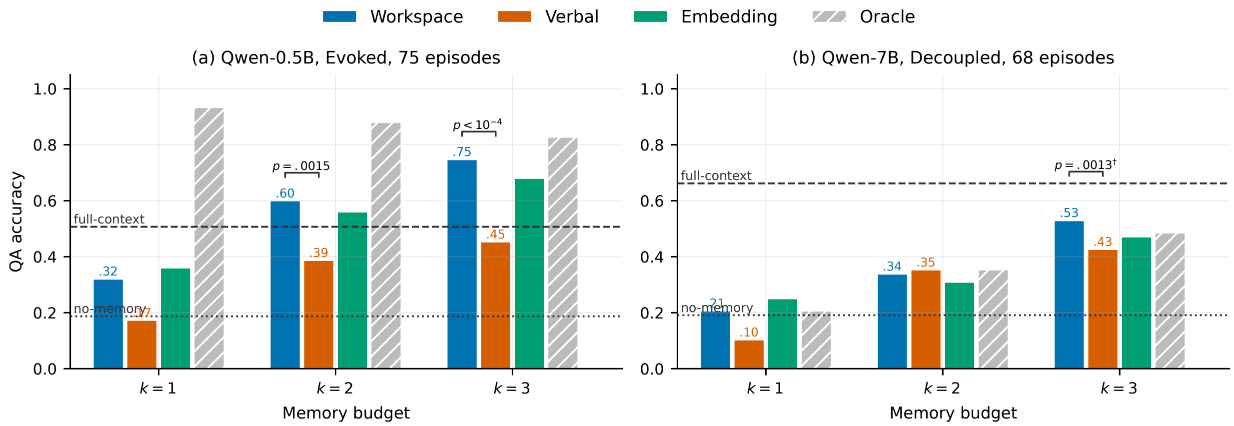

Figure 4: Budget-limited recall QA. Left: 0.5B on the Evoked benchmark, where workspace gating beats verbal gating (k=2: $+20/-4$, p=0.0015; k=3: $+24/-2$, $p<10^{-4}$), which sits at the no-memory floor. Right: 7B on the construct-valid Decoupled benchmark (64-token answers), where workspace gating matches the label oracle and beats verbal gating (k=3, †: $+26/-7$, p=0.0013, computed on the 120-episode scaled benchmark). The two panels are the two regimes' primary cells; the full grid, which also contains null and reversed cells at these n's, and all McNemar tests are in App. C.

Table 2: Causal effect of workspace interventions on memory-based answers. Steering the residual stream along the stored bridge's input-side direction (real) vs. an unrelated, norm-matched direction (sham) while the model answers from memory; Qwen2.5 checkpoints; 20 disjoint episode pairs; a flip is a sign change of the stored-vs-target answer log-odds. Exact McNemar on paired real-vs-sham flips; exact two-sided Wilcoxon signed-rank on paired Δlog-odds. Specificity (real high, sham low) holds at gentle steering scales and degrades as strength grows.

Model	Scale	Real flip	Sham flip	McNemar	p	Wilcoxon p
7B	0.5	0.70	0.05	$+13/-0$	0.0002	4.8e-05
7B	1.0	0.90	0.30	$+12/-0$	0.0005	1.9e-06
3B	0.5	0.60	0.10	$+11/-1$	0.0063	1.3e-04
3B	1.0	0.85	0.35	$+10/-0$	0.0019	3.8e-06
0.5B	1.0	0.75	0.20	$+11/-0$	0.0010	1.7e-04

gating reaches 0.75 at k=3; the paired advantage is significant despite only 75 episodes. Budget-k payoff tracks within-episode ranking rather than the pooled AUC of Eq. 3 (top-3 containment 0.707 here; App. C). The canonical 1–10 rating variant is a stronger verbal baseline on this answer-coupled benchmark (matching workspace there) but sits at chance predictively on the construct-valid Decoupled benchmark (App. J). At 7B on the 120-episode construct-valid benchmark (Decoupled-L), workspace gating shows no detectable difference from the selection oracle (0.525 vs. 0.525 at k=3; paired difference 0.000, 95% CI $[-0.092, +0.083]$; workspace nominally exceeds the oracle in two other cells, indicating the oracle itself is noise-limited at this n) and beats verbal gating on the same set ($+26/-7$, p=0.0013). Recency, random, and the surface-feature fusion proxy lose to workspace gating throughout (App. C, K); embedding gating is the strongest alternative but never separates from workspace downstream and is predictively dispatched on Decoupled. Two boundary conditions: at 3B on Decoupled-class tasks the recall-time composition step, not selection, is the binding constraint (oracle 0.43 vs. full-context 0.57), and with a 24-token answer budget even the oracle collapses: when the model cannot exploit a stored item at recall time, selection quality has no effect.

5.4 Causal Analysis

The predictive results alone do not establish that the answer computation uses the representations the readout measures. To test this directly, we steer the residual stream along the stored bridge's input-side difference-of-means direction (Zou et al., 2023) toward a different bridge while the model answers from memory, versus sham directions built from unrelated bridges on held-out episodes (20 disjoint pairs). Table 2: at the gentlest scale the real direction flips the answer in 0.60 of pairs versus 0.10 for sham at 3B (McNemar $+11/-1$, p=0.0063) and in 0.70 versus 0.05 at 7B ($+13/-0$, $p=2.4\times10^{-4}$), directionally consistent at 0.5B (tested at scale 1.0). A flip means the answer log-odds

Table 3: Write-admission policies on the mixed-provenance pool (Explicit ∪ Evoked ∪ Decoupled-L; 283 episodes; per-benchmark generation budgets fixed across policies; all cells from a single run generation). Bold: column best among content-scoring policies; the two internal signals (likelihood, workspace) are statistically indistinguishable in every cell (all $p \geq 0.10$) and each beats the binary probe. The absence heuristic exploits the single-planted-inference construction and collapses on multi-absent pools (three-act analysis, App. P); pairwise tests and the 1–10 rating analysis are also there.

	0.5B		7B	
Policy	k=2	k=3	k=2	k=3
Verbal (binary probe)	0.290	0.329	0.463	0.555
Verbal (1–10 rating)	0.343	0.410	0.491	0.601
Embedding	0.350	0.442	0.477	0.583
Cond. likelihood (one pass/cand.)	**0.375**	**0.466**	**0.537**	0.618
Workspace (zero passes)	0.332	0.431	0.502	**0.636**
PRG (routed)	0.336	0.417	**0.512**	0.565
Absence heuristic (diagnostic)	0.459	0.459	0.629	0.636
Oracle	0.512	0.484	0.657	0.707

between the stored and the steered-toward bridge change sign; sham directions are built from unrelated bridges of held-out episodes at matched norm. At stronger steering scales real flips rise further but sham flips rise too (App. G), so the gentle-scale contrast carries the specificity claim. Output-side (unembedding) steering has no effect, consistent with Anthropic Interpretability Team (2026). The concept representations the readout scores are therefore causally used by memory-based answering; steering the unembedding rows (the output side) does nothing, so the readout taps content, not output preparation.

5.5 Gating Evaluation

Table 3 evaluates the gate in the deployment-realistic setting: a pooled stream mixing all three provenance regimes, where any channel that fails on one regime pays for it. Because the contribution is an admission score rather than a memory system, the comparison holds the pipeline fixed and varies only the score; an end-to-end comparison against full memory frameworks would confound retrieval, organization, and prompting with the admission signal under study, and every such framework can adopt whichever score wins here. Three results follow. First, replacing the binary importance probe with the workspace readout wins at k=3 at both scales ($+47/-18$, p=0.0004, confirmatory under Holm correction over the full test family; $+45/-22$, p=0.0067 uncorrected, marginal after correction; App. P), and the workspace channel is never significantly worse than any content-scoring policy in any cell. Second, the failure is specific to the verbal channel, not to internal signals generally: against the canonical 1–10 rating, the strongest verbal form, workspace gating is numerically ahead at 7B (0.636 vs. 0.601, p=0.31) at zero probe cost, where the rating spends one forward pass per candidate, and on the construct-valid slice the rating is predictively at or below chance (App. J). An internal conditional-likelihood score at that same one-pass cost gates indistinguishably from the workspace readout (all cells $p \geq 0.10$) and itself beats the binary probe ($p \leq 0.035$ in three of four cells, p=0.051 in the fourth; App. P); the readout is the internal signal that needs no extra passes and carries the causal grounding of §5.4. Third, at the tight 7B budget provenance routing is the strongest policy that spends no per-candidate forward passes (0.512 vs. 0.502 single-channel) and helps where the channels are complementary; the single-channel gate is the recommended default at capable scale. Calibration matters: z-normalized routing loses admission slots for inferred items systematically because the two score distributions differ in shape, and underperforms rank calibration throughout (App. P). Finally, because each benchmark plants a single inference, bare absence is nearly an oracle proxy here; App. P analyzes this diagnostic and the multi-absent stress tests, where the degenerate absence heuristic collapses and, when the competing absent candidates are semantically confusable, the workspace readout is the leading channel: its selection-signal advantage over the embedding replicates on a fresh 120-episode set (paired ΔAUC +0.064, one-sided p=0.026; pooled estimate +0.058 [+0.007, +0.108]) and gated QA shows no detectable difference from the true-label oracle (App. P). On the two standard naturalistic streams (LoCoMo, LongMemEval), whose candidates are surface content by construction, no question-blind channel, ours or the verbal importance probe, separates from random under LLM-judge grading: the construct that separates the channels is structurally absent there, which is precisely why the controlled benchmarks of §3.2 are needed. When that construct is planted back, by embedding our Decoupled episodes inside real LoCoMo sessions, the dissociation reappears at 7B with the workspace readout unattenuated ($W - V$ +0.21

Table 4: Metacognitive alignment, two model families: ∼500 fine-tuning steps on yes/no labels derived from each model's own workspace ranking, using disjoint benchmarks (Evoked, Explicit, Evoked-G2). The verbal channel is repaired on fully held-out benchmarks (+0.28–0.39 AUC) while the workspace channel is untouched, full-context QA is statistically unchanged, and general capability (MMLU, GSM8K, ARC) is unaffected (Table 17) (App. L). For the Qwen-0.5B Compositional row the repaired reporter converges to its teacher's (chance) level; where generation-time computation adds signal (OLMo Compositional) the student can exceed the static teacher (§6.1).

		Verbal report V (AUC)		Workspace W	
Model	Held-out benchmark	before	after	before	after
OLMo-2-1B	Decoupled	0.326 [0.251, 0.400]	**0.671** [0.584, 0.751]	0.580	0.586
OLMo-2-1B	Compositional	0.267 [0.179, 0.360]	**0.580** [0.482, 0.677]	0.477	0.491
Qwen2.5-0.5B	Decoupled	0.281 [0.210, 0.354]	**0.668** [0.595, 0.737]	0.526	0.524
Qwen2.5-0.5B	Compositional	0.172 [0.112, 0.241]	**0.451** [0.367, 0.534]	0.416	0.409

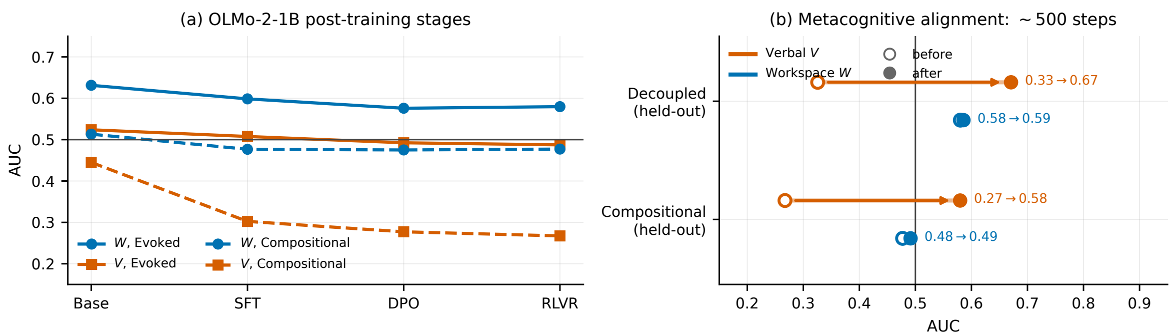

Figure 5: Post-training changes the verbal channel, not the workspace channel, and the verbal channel is repairable. (a) OLMo-2-1B released post-training stages: the workspace channel (W, blue) never improves from base, on either benchmark, while the verbal channel (V, orange) stays at chance (Evoked) or far below it (Compositional). **(b)** Metacognitive alignment: ∼500 self-distillation steps move the verbal channel (arrows) from below-chance to well-calibrated on fully held-out benchmarks while leaving the workspace channel in place (open circle = before, filled = after).

$[+0.10, +0.31]$): the effect is carried by content provenance, not by synthetic style (App. N).

6 Metacognitive Alignment and Availability Readouts

6.1 Metacognitive Alignment

If the model knows more than it tells because telling was never trained against the knowing (Fig. 5), the reporter should be trainable from the model's own workspace signal, without human annotation. We fine-tune OLMo-2-1B-Instruct for ∼500 steps on the exact importance-probe prompt, with yes/no targets derived from the model's own workspace ranking (top-2 items per episode), using only the explicit and evoked benchmarks plus the second-generator benchmark (Evoked-G2, gpt-4o); the decoupled benchmarks (Decoupled, Compositional) are never seen. Table 4 (seed-0 runs; three-seed means 0.62–0.65, App. L): held-out verbal calibration jumps from 0.326 to 0.671 and from 0.267 to 0.580. The same procedure, unchanged, replicates in a second family: Qwen2.5-0.5B-Instruct rises from 0.281 to 0.668 (Decoupled) and 0.172 to 0.451 (Compositional). Three controls secure the interpretation: label-shuffled fine-tunes leave held-out calibration at chance in both families (0.416/0.382), distilling from embedding-derived labels fails (0.458), and the aligned reporter rejects context-absent foils (0.900), so the gain is specifically the workspace signal, not format de-biasing, generic salience, or an absence shortcut (App. L). The two Compositional endpoints then fit one prediction: distillation transfers the monitor's signal, so where the static teacher is at chance (Qwen-0.5B) the student plateaus there, while where generation-time computation adds signal beyond the static snapshot (OLMo-1B) the student can exceed the static value, behaving like the dynamic readouts of §6.2. The calibration transfers across held-out benchmarks and generators, which is consistent with the reporter acquiring a benchmark-independent readout of internal state; shared construction biases across generators cannot be fully excluded. The workspace channel itself

Table 5: Availability readouts by provenance regime (Qwen-7B AUC; full grids incl. CIs in App. F). Decode-based members win on evoked content; the utility-gradient member is the only family member whose paired gain over the static readout is significant on Compositional ($W_J - W_{rr}$: $+0.125$ [$+0.010$, $+0.244$]; head-to-head vs. the conditional-likelihood baseline it does not separate, App. J); replay's Compositional ceiling is budget-independent. Bold: column best.

Member	Evoked	Decoupled	Compositional
Spotlight W (static)	0.633	**0.641**	0.514
Broadcast breadth (static)	0.637	0.557	0.486
Rehearsal W_{rep} (dynamic)	**0.684**	0.578	0.553
Leverage W_J (dynamic)	0.506	0.550	**0.639**

is untouched, and full-context QA never moves significantly (-0.029 for OLMo, $+0/-2$, p=0.50; -0.044 for Qwen, $+5/-8$, p=0.58; App. L). Practically, twenty minutes of self-supervised fine-tuning on one GPU repairs a channel that SFT, DPO, and RLVR left miscalibrated.

6.2 A Family of Availability Readouts

Availability to downstream computation can be operationalized in more than one way. We evaluate four readouts of the same frozen model: *Spotlight* (W, instantaneous decodability), *Broadcast* (its spatial spread across positions, in the spirit of global-workspace ignition, Dehaene et al., 2011), *Rehearsal* (W_{rep}: reactivation across K sampled probe-blind continuations, an analogue of offline replay), and Leverage (W_J: the directional derivative of expected future answering utility along the concept's input-side direction, the infinitesimal form of the causal intervention in §5.4, computed by backpropagation without training: one forward-plus-backward pass per episode–question pair, with every candidate scored from the same gradient). Table 5 shows a clean double dissociation between readouts: decode-based members own the evoked regime (Rehearsal reaches 0.684 on Evoked, improving on Spotlight), while Leverage is the only member that recovers compositional bridges under the paired criterion of §5.2 (Compositional at 7B: 0.639; paired over Spotlight $+0.125$ $[+0.010, +0.244]$, CI excluding zero), at the cost of low sensitivity where decoding is easy. Rehearsal's Compositional ceiling (≈ 0.55) does not move with rollout budget: free-running generation rarely performs the composition spontaneously, whereas the gradient detects the latent structure without requiring it to run. Boundaries: the Leverage result does not replicate on OLMo-2-7B (0.515); head-to-head against the conditional-likelihood baseline on Compositional the two do not separate (W_J−LL $+0.046$ $[-0.056, +0.153]$, App. J), so the claim is that Leverage is the only *workspace* readout whose paired gain over the static member is significant, not that it beats every non-workspace score; and naive member ensembles do not beat the per-regime best member (App. F); availability estimation is regime- and family-dependent, and the dissociation between readouts parallels, at a lower level, the report–computation dissociation that motivates this work.

7 Conclusion

Agent memory systems inherit a convenient assumption: that a model can say what is worth remembering. Under controlled content provenance the assumption breaks in a structured way: self-rated importance is excellent for explicit content and, on the construct-valid benchmarks, at or below chance for silently inferred content at every scale and post-training stage we measured, in three model families, with the cue-density boundary mapped in App. I. A J-lens-derived workspace readout recovers much of the signal that self-report loses on evoked content and causally feeds memory-based answers; used as a write gate (with PRG, a provenance-routed extension, for tight budgets on mixed streams), it matches the canonical 1–10 rating at zero probe cost while leading on the construct-valid slice, selects a memory budget at up to selection-oracle quality, and beats the binary importance probe on a mixed-provenance stream. The failure is remediable: post-training never connects the verbal channel to the workspace because none of its objectives depend on that connection, and roughly 500 self-supervised steps suffice to establish it. Beyond agents, we read this as a post-training lesson: report-channel objectives do not by themselves calibrate self-report against internal state, but self-distillation from internal signals cheaply can.

Limitations. Our benchmarks are synthetic and single-word-concept by design; control over provenance is the point, and it is not available elsewhere: on the two standard naturalistic benchmarks (LoCoMo, LongMemEval) every candidate is surface content by construction, no question-blind channel separates from random there, and the evidence oracle doubles all of them (App. N). A naturalistic stream rich in silent inference does not yet exist and is the natural next instrument; embedding our episodes in real dialogue carriers preserves the dissociation at 7B (App. N), so the obstacle is benchmark construction, not transfer of the effect. Each benchmark is written by one LLM generator, so we replicated across three: the workspace readout is stable across the generators that produce the construct-valid design (0.53–0.66; on gpt-4o's weakly evoked bridges both channels attenuate), while the verbal channel's calibration on silent inferences swings in both directions, anti-calibrated on sparse-cue episodes at every scale and recovering on densely cued ones at 7B (App. I); the anti-calibration claim is scoped accordingly, and the workspace channel's stability is precisely its case. Episode counts (52–120 per benchmark) give adequate power only for the pre-specified contrasts. The Leverage readout's compositional-regime win holds for Qwen but not OLMo-2-7B; the 3B downstream gap on Decoupled-class tasks is bounded by recall-time composition, not selection; and in vision–language probes, visually presented identity behaves like the explicit regime (near-perfect ask-time verbal access in both Qwen2-VL-2B and LLaVA-1.5-7B), while its word-level workspace trace is family-dependent, absent at 2B and partial in LLaVA-7B (App. H), so our claims are about language-derived inference. Post-training claims are scoped to our stage sweep and to utility-salience. Our results establish the value of training-free internal readouts for write admission; whether supervised hidden-state probes do better is open (they would require labeled training data that the write-gate setting does not naturally provide). Full deviations from the pre-specified predictions are listed in App. O.

Outlook. Three extensions look immediate: widening PRG's routing to the full availability family (the gradient-based Leverage readout for compositional content), metacognitive alignment at scale and for other self-knowledge channels, and evaluating workspace gating inside production memory stacks on naturalistic long-horizon streams.

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A Benchmark Construction and Validation

Generation. All released benchmarks are generated by `gpt-4.1`; the replication generators are `gpt-4o` (Evoked design) and `gpt-5.6` (Decoupled design: $78/80$ raw episodes pass the same mechanical validation, 42 episodes / 217 items after concept deduplication; its episodes carry several convergent cues per bridge). Generation proceeds in batches of five episodes under the hard rules of §3, then mechanically validated and filtered. Validation rules and yields: **Explicit**: concepts must appear in text; merged 140 raw episodes → 88 after global concept deduplication (417 items, 131 load-bearing). **Evoked**: the load-bearing bridge must not appear as a substring of the context; 132 raw → 75 episodes (352 items, 75 positives). **Decoupled**: additionally the gold answer must differ from every stored concept and be absent from the context, and the probe may share no content word with the context except the protagonist's name; 220 raw → 68 episodes (335 items), scaled version 300 raw → 120 episodes (572 items), person names deduplicated. `gpt-4o` satisfies these constraints in only $4/150$ attempts, which is why the independent-generator replication uses the Evoked design (57 episodes after dedup). **Compositional**: bridge silent and answer decoupled but probes may reference context anchors; 80 raw → 52 episodes (261 items).

Leakage audits. A did-the-word-appear classifier is at chance on Explicit (AUC 0.48; $\sim 98\%$ of concepts appear literally in both classes); Evoked–Decoupled remove the confound by construction. A full-lexeme scan of Decoupled finds 2/68 bridges sharing a lexeme with the context (e.g. *yankees*/"Yankee Stadium"), 1/68 answers as an inflection of a context word, and 7/68 probes sharing one generic inflected content word; dropping all 12 flagged episodes leaves every statistical conclusion unchanged (7B $W \approx 0.63$ with or without the flagged episodes; all $W-V$ CIs exclude 0). Compositional's probes carry surface anchors, which makes Compositional valid for encoding-time prediction but invalid for downstream payoff (the no-memory floor reaches 0.40 at 7B); Decoupled was constructed to close exactly this, and the 24-token oracle collapse (Table 7) motivated the 64-token answer budget for Decoupled-class downstream runs.

B Full Predictive Grids

Table 6: Full W / V AUC grid, all measured checkpoints × final batteries. Parenthesized V values are the template-free probe V_{raw} (base models). "—": not measured (see App. M for the run manifest).

Checkpoint	Explicit	Evoked	Decoupled	Compositional
Qwen2.5-0.5B	0.522 / (0.508)	0.538 / (0.259)	—	0.430 / (0.126)
Qwen2.5-0.5B-I	0.521 / 0.551	0.516 / 0.295	0.526 / 0.281	0.416 / 0.172
Qwen2.5-1.5B	0.509 / (0.589)	0.582 / (0.588)	—	0.544 / (0.381)
Qwen2.5-1.5B-I	0.513 / 0.615	0.579 / 0.522	0.577 / 0.462	0.549 / 0.290
Qwen2.5-3B	0.538 / (0.608)	0.616 / (0.574)	—	0.521 / (0.329)
Qwen2.5-3B-I	0.533 / 0.641	0.579 / 0.461	0.594 / 0.254	0.498 / 0.197
Qwen2.5-7B	0.543 / (0.592)	0.621 / (0.348)	—	0.516 / (0.110)
Qwen2.5-7B-I	0.538 / 0.681	0.633 / 0.579	0.641 / 0.486	0.514 / 0.344
GPT-2	0.493 / (0.536)	0.540 / (0.539)	—	0.551 / (0.592)
OLMo-2-1B (RLVR)	0.536 / 0.557	0.580 / 0.487	0.580 / 0.326	0.477 / 0.267
OLMo-2-7B	—	0.673 / (0.493)	0.684 / (0.317)	—
OLMo-2-7B-I	0.492 / 0.622	0.669 / 0.539	0.665 / 0.352	0.534 / 0.296
Mistral-7B-I	0.532 / 0.622	0.670 / 0.589	0.672 / 0.481	0.596 / 0.322

Table 6 gives the full checkpoint grid. Episode-cluster bootstrap CIs (B=4,000) for all headline cells, and Bonferroni-corrected primary contrasts, are included in the released statistical tables; item-level and cluster CIs never disagree on any significance call reported in the main text except 1.5B-Decoupled $W-V$ under the raw-probe variant, where the item-level CI excludes zero and the episode-cluster CI does not; we treat the cluster CI as authoritative there.

C Downstream: All Runs and All Tests

Table 7 lists all budget-limited runs and Table 8 every McNemar test; the tests outside the pre-specified contrasts are uncorrected, and they include one nominal reversal (verbal over workspace on Evoked at 7B, k=1: $+6/-18$, p=0.023) alongside the workspace-favoring cells quoted in the main text.

Within-episode selection metrics. The pooled AUC of Eq. 3 compares items across episodes, while gating ranks within an episode, and the two can diverge when score scales differ across episodes. For the two downstream headline cells: Evoked 0.5B has pooled workspace AUC 0.516 but within-episode AUC 0.537 and top-3 load-bearing containment 0.707, which is what drives recall-QA 0.75; Decoupled-L 7B has pooled 0.599, within-episode 0.623, and containment 0.783. Budget-k payoff tracks the within-episode quantities.

Table 7: Budget-limited recall QA, all runs. Each cell shows accuracy at $k{=}1/2/3$ (leading zeros omitted); generation budget as marked per row; random = 3-seed mean.

Benchmark	Size	workspace	verbal	embedding	recency	random	oracle	no-mem	full-ctx
Evoked (24t)	0.5B	.32/.60/.75	.17/.39/.45	.36/.56/.68	.19/.17/.25	.21/.37/.53	.93/.88/.83	.19	.51
Evoked (24t)	1.5B	.28/.56/.65	.29/.44/.57	.35/.55/.64	.07/.16/.27	.12/.32/.51	.79/.80/.81	.20	.69
Evoked (24t)	3B	.17/.49/.63	.21/.39/.51	.24/.43/.63	.03/.11/.20	.08/.24/.42	.72/.69/.81	.04	.68
Evoked (24t)	7B	.32/.63/.87	.48/.65/.76	.40/.61/.76	.09/.21/.31	.12/.38/.59	.92/.97/.99	.17	.71
Decoupled (24t)	0.5B	.12/.19/.19	.10/.15/.16	.07/.18/.27	.12/.15/.19	.11/.13/.19	.16/.18/.21	.15	.35
Decoupled (24t)	1.5B	.23/.31/.28	.25/.31/.27	.31/.34/.38	.13/.18/.25	.16/.28/.25	.27/.31/.32	.22	.53
Decoupled (24t)	3B	.03/.04/.09	.03/.03/.09	.03/.06/.10	.03/.06/.13	.02/.06/.12	.07/.07/.15	.04	.47
Decoupled (24t)	7B	.15/.21/.22	.10/.21/.21	.22/.25/.28	.07/.13/.28	.10/.19/.24	.13/.21/.27	.19	.62
Decoupled (64t)	3B	.13/.25/.34	.07/.18/.32	.15/.25/.34	.10/.22/.32	.10/.22/.31	.29/.32/.46	.13	.54
Decoupled (64t)	7B	.21/.34/.53	.10/.35/.43	.25/.31/.47	.07/.19/.41	.13/.30/.36	.21/.35/.48	.19	.66
Decoupled-L (64t)	0.5B	.15/.22/.26	.14/.20/.22	.14/.23/.33	.15/.17/.23	.15/.19/.23	.25/.28/.30	.20	.35
Decoupled-L (64t)	1.5B	.24/.32/.35	.22/.33/.33	.28/.35/.39	.17/.21/.26	.16/.28/.34	.33/.37/.38	.23	.50
Decoupled-L (64t)	3B	.16/.25/.36	.13/.19/.33	.15/.23/.31	.11/.18/.28	.09/.21/.33	.32/.36/.43	.10	.57
Decoupled-L (64t)	7B	.23/.42/.53	.13/.30/.37	.23/.36/.45	.07/.23/.38	.14/.33/.40	.33/.42/.53	.17	.62

Table 8: Exact McNemar tests, workspace vs. each rival policy on identical episodes; cells show discordant counts $+a/-b$ (episodes only workspace / only rival correct) with exact p in parentheses.

Benchmark	Size	Rival	$k{=}1$	$k{=}2$	$k{=}3$
Evoked (24t)	0.5B	embedding	$+11/-14$ (0.690)	$+13/-10$ (0.678)	$+11/-6$ (0.332)
		oracle	$+0/-46$ $(<10^{-3})$	$+1/-22$ $(<10^{-3})$	$+7/-13$ (0.263)
		recency	$+14/-4$ (0.031)	$+34/-2$ $(<10^{-3})$	$+37/-0$ $(<10^{-3})$
		verbal	$+16/-5$ (0.027)	$+20/-4$ (0.002)	$+24/-2$ $(<10^{-3})$
Evoked (24t)	1.5B	embedding	$+9/-14$ (0.405)	$+14/-13$ (1.000)	$+12/-11$ (1.000)
		oracle	$+0/-38$ $(<10^{-3})$	$+2/-20$ $(<10^{-3})$	$+2/-14$ (0.004)
		recency	$+17/-1$ $(<10^{-3})$	$+32/-2$ $(<10^{-3})$	$+31/-2$ $(<10^{-3})$
		verbal	$+11/-12$ (1.000)	$+22/-13$ (0.175)	$+16/-10$ (0.327)
Evoked (24t)	3B	embedding	$+5/-10$ (0.302)	$+18/-13$ (0.473)	$+10/-10$ (1.000)
		oracle	$+1/-42$ $(<10^{-3})$	$+5/-20$ (0.004)	$+4/-18$ (0.004)
		recency	$+11/-0$ (0.001)	$+33/-4$ $(<10^{-3})$	$+34/-2$ $(<10^{-3})$
		verbal	$+4/-7$ (0.549)	$+22/-14$ (0.243)	$+18/-9$ (0.122)
Evoked (24t)	7B	embedding	$+12/-18$ (0.362)	$+14/-13$ (1.000)	$+11/-3$ (0.057)
		oracle	$+0/-45$ $(<10^{-3})$	$+0/-26$ $(<10^{-3})$	$+0/-9$ (0.004)
		recency	$+18/-1$ $(<10^{-3})$	$+33/-2$ $(<10^{-3})$	$+42/-0$ $(<10^{-3})$
		verbal	$+6/-18$ (0.023)	$+12/-14$ (0.845)	$+14/-6$ (0.115)
Decoupled (24t)	0.5B	embedding	$+5/-2$ (0.453)	$+6/-5$ (1.000)	$+6/-11$ (0.332)
		oracle	$+1/-4$ (0.375)	$+4/-3$ (1.000)	$+5/-6$ (1.000)
		recency	$+5/-5$ (1.000)	$+10/-7$ (0.629)	$+6/-6$ (1.000)
		verbal	$+5/-4$ (1.000)	$+8/-5$ (0.581)	$+9/-7$ (0.804)
Decoupled (24t)	1.5B	embedding	$+3/-8$ (0.227)	$+3/-5$ (0.727)	$+7/-14$ (0.189)
		oracle	$+4/-6$ (0.754)	$+6/-6$ (1.000)	$+5/-8$ (0.581)
		recency	$+10/-3$ (0.092)	$+11/-2$ (0.022)	$+9/-7$ (0.804)
		verbal	$+4/-5$ (1.000)	$+6/-6$ (1.000)	$+9/-8$ (1.000)
Decoupled (24t)	3B	embedding	$+1/-1$ (1.000)	$+2/-3$ (1.000)	$+5/-6$ (1.000)
		oracle	$+0/-3$ (0.250)	$+1/-3$ (0.625)	$+4/-8$ (0.388)
		recency	$+2/-2$ (1.000)	$+1/-2$ (1.000)	$+2/-5$ (0.453)
		verbal	$+2/-2$ (1.000)	$+2/-1$ (1.000)	$+5/-5$ (1.000)
Decoupled (24t)	7B	embedding	$+3/-8$ (0.227)	$+5/-8$ (0.581)	$+4/-8$ (0.388)
		oracle	$+4/-3$ (1.000)	$+6/-6$ (1.000)	$+5/-8$ (0.581)
		recency	$+7/-2$ (0.180)	$+8/-3$ (0.227)	$+7/-11$ (0.481)
		verbal	$+5/-2$ (0.453)	$+7/-7$ (1.000)	$+7/-6$ (1.000)
Decoupled (64t)	3B	embedding	$+3/-4$ (1.000)	$+8/-8$ (1.000)	$+10/-10$ (1.000)
		oracle	$+1/-12$ (0.003)	$+5/-10$ (0.302)	$+3/-11$ (0.057)
		recency	$+7/-5$ (0.774)	$+10/-8$ (0.815)	$+9/-8$ (1.000)
		verbal	$+7/-3$ (0.344)	$+10/-5$ (0.302)	$+11/-10$ (1.000)
Decoupled (64t)	7B	embedding	$+6/-9$ (0.607)	$+13/-11$ (0.839)	$+12/-8$ (0.503)
		oracle	$+5/-5$ (1.000)	$+9/-10$ (1.000)	$+11/-8$ (0.648)
		recency	$+11/-2$ (0.022)	$+15/-5$ (0.041)	$+16/-8$ (0.152)
		verbal	$+9/-2$ (0.065)	$+10/-11$ (1.000)	$+15/-8$ (0.210)
Decoupled-L (64t)	0.5B	embedding	$+4/-3$ (1.000)	$+14/-15$ (1.000)	$+7/-16$ (0.093)
		oracle	$+1/-13$ (0.002)	$+7/-14$ (0.189)	$+8/-13$ (0.383)
		recency	$+5/-5$ (1.000)	$+14/-9$ (0.405)	$+10/-6$ (0.455)
		verbal	$+4/-3$ (1.000)	$+12/-10$ (0.832)	$+12/-7$ (0.359)
Decoupled-L (64t)	1.5B	embedding	$+7/-12$ (0.359)	$+10/-14$ (0.541)	$+16/-21$ (0.511)

Benchmark	Size	Rival	k=1	k=2	k=3
		oracle	$+8/-18$ (0.075)	$+11/-17$ (0.345)	$+12/-15$ (0.701)
		recency	$+14/-5$ (0.064)	$+21/-8$ (0.024)	$+23/-12$ (0.089)
		verbal	$+11/-8$ (0.648)	$+13/-15$ (0.851)	$+20/-17$ (0.743)
Decoupled-L (64t)	3B	embedding	$+7/-6$ (1.000)	$+14/-12$ (0.845)	$+17/-11$ (0.345)
		oracle	$+4/-23$ ($<10^{-3}$)	$+11/-24$ (0.041)	$+9/-18$ (0.122)
		recency	$+14/-8$ (0.286)	$+21/-13$ (0.230)	$+21/-12$ (0.163)
		verbal	$+10/-7$ (0.629)	$+15/-8$ (0.210)	$+18/-14$ (0.597)
Decoupled-L (64t)	7B	embedding	$+11/-10$ (1.000)	$+26/-18$ (0.291)	$+22/-13$ (0.175)
		oracle	$+9/-20$ (0.061)	$+16/-15$ (1.000)	$+14/-14$ (1.000)
		recency	$+22/-3$ ($<10^{-3}$)	$+33/-9$ ($<10^{-3}$)	$+28/-10$ (0.005)
		verbal	$+14/-2$ (0.004)	$+25/-10$ (0.017)	$+26/-7$ (0.001)

D Post-Training Stages

Table 9: OLMo-2-0425-1B post-training stages. The workspace channel never improves (base is highest on Evoked); the verbal channel is miscalibrated already in the raw-probed base model and becomes more confidently wrong through the chat template.

Benchmark	Stage	W [95% CI]	V (chat)	V_{raw}
Evoked	BASE	0.631 [0.565, 0.696]	—	0.524
	SFT	0.599 [0.528, 0.665]	0.508	0.484
	DPO	0.576 [0.502, 0.644]	0.492	0.498
	RLVR	0.580 [0.506, 0.648]	0.487	0.510
Compositional	BASE	0.513 [0.421, 0.611]	—	0.445
	SFT	0.477 [0.384, 0.573]	0.302	0.313
	DPO	0.475 [0.380, 0.577]	0.277	0.314
	RLVR	0.477 [0.381, 0.576]	0.267	0.331

Table 9 lists the OLMo-2 stage sweep. Qwen2.5 base→instruct paired differences on the Evoked benchmark (same items, paired bootstrap): 0.5B -0.022 (P=0.038, a degradation); 1.5B -0.004; 3B -0.037 $[-0.065, -0.011]$ (significant degradation); and 7B $+0.012$ (n.s.). On Compositional: -0.013, $+0.005$, -0.023, and -0.002 (all n.s.). OLMo-2 transitions: base→SFT -0.033 (P=0.116); SFT→DPO -0.023 $[-0.042, -0.005]$; DPO→RLVR $+0.004$ (n.s.); cumulative base→RLVR -0.052 (one-sided episode-cluster bootstrap P=0.041).

E FutureLens Negative Control

Table 10 gives all FutureLens cells.

F The Availability Family: Full Grids

Definitions. Rehearsal: K=8 rollouts of 40 tokens at T=0.7 (probe-blind, raw context); W_{rep}^{emit} = fraction of rollouts emitting a case-variant of the concept; W_{rep}^{dec} = mean over rollouts of the peak reciprocal rank across continuation positions and the six layers at fractional depths $\{0.30, 0.45, 0.60, 0.75, 0.90, 1.0\}$. Doubling the budget ($K$=16, 64 tokens) does not lift Compositional (0.532/0.560). Leverage:

$$W_J \;=\; \max_{\ell \in \mathcal{B}} \; \cos\!\big(\mathbb{E}_q\big[\nabla_{h_\ell(x)} \log p_{\mathcal{M}}(\hat{a}_q \mid x, q)\big],\; \hat{v}_c^{(\ell)}\big),$$

where the gradient is taken at the end-of-context position of $h_\ell(x)$ and $\mathcal{B}$ contains the four layers at fractional depths 0.30, 0.45, 0.60, and 0.75. The expectation runs over a fixed 12-question probe-blind bank; $\hat{v}_c^{(\ell)}$ is the concept's input-side difference-of-means direction at layer ℓ; $\hat{a}_q$ is the model's own greedy answer, held fixed while differentiating (pathwise gradient). This is the $\varepsilon \to 0$ tangent of the Table 2 intervention. Ignition: breadth $\times$ $\sqrt{\text{persistence}}$ $\times$ $(1+\max(0, \text{sharpness}))$ over the (layer $\times$ position) reciprocal-rank grid of the last half of the context (breadth = fraction of positions whose peak reciprocal rank over late layers exceeds the top-k threshold; persistence = grid mean; sharpness = largest layer-to-layer increase of the position-max profile). Naive rank-average or max ensembles of members do not exceed the per-regime best member (Compositional 7B: 0.594/0.631 vs. Leverage alone 0.639).

Table 10: Trained tuned-future-lens negative control: initialized at the model's own unembedding, trained on disjoint Explicit-benchmark contexts to place mass on a 12-token future window. It never rescues Compositional-benchmark bridges and is significantly *worse* than the logit lens on evoked bridges at 7B; the static-state boundary is not a readout artifact.

Benchmark	Size	FutureLens AUC	Logit-lens W	FL$-W$ [95% CI]
Compositional	0.5B	0.523 [0.432, 0.611]	0.416	+0.107 [+0.029, +0.189]*
Compositional	1.5B	0.585 [0.496, 0.673]	0.549	+0.035 [-0.034, +0.113]
Compositional	3B	0.525 [0.435, 0.615]	0.498	+0.028 [-0.052, +0.112]
Compositional	7B	0.563 [0.473, 0.658]	0.514	+0.049 [-0.053, +0.149]
Evoked	0.5B	0.507 [0.435, 0.575]	0.516	-0.009 [-0.081, +0.065]
Evoked	1.5B	0.562 [0.495, 0.632]	0.579	-0.016 [-0.091, +0.058]
Evoked	3B	0.576 [0.511, 0.643]	0.579	-0.003 [-0.071, +0.068]
Evoked	7B	0.545 [0.476, 0.614]	0.633	-0.088 [-0.157, -0.022]*

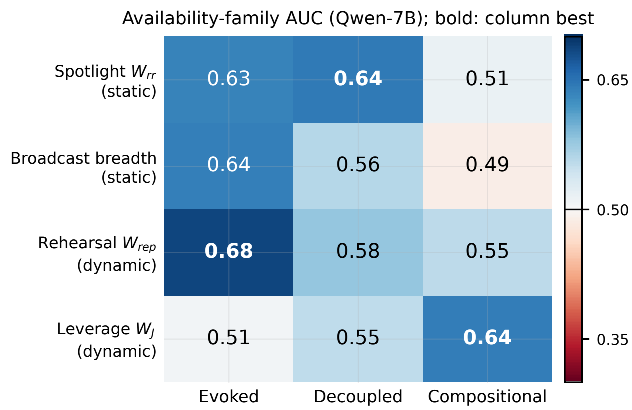

Figure 6: The availability-family double dissociation as a matrix (Qwen-7B): decode-based members (rows 1–3) own the Evoked regime; the utility-gradient member (row 4) is the only readout with a significant paired gain over the static readout on the Compositional benchmark (Table 5); the conditional-likelihood baseline's nominal edge there does not reach significance (App. J).

G Causal Intervention: Full Results

Procedure. Concept directions are input-side difference-of-means vectors (mean residual state over $\sim$20 contexts naming the concept minus matched negatives), estimated per layer in the causal band ($n/4$ to $n/4+n/3$ of depth); steering adds the direction, scaled as a fraction of the local hidden-state norm, at every band layer and position while the model answers from memory; sham directions are built identically from bridges of held-out episodes and norm-matched. Pairs consume disjoint episodes from the front of the benchmark.

Independent-pairs results are in Table 2, including the 7B run (gentle scale 0.5: flip 0.70 vs. sham 0.05, $+13/-0$, p=2.4×10^{-4}; scale 1.0: 0.90 vs. 0.30, specificity again degrading with strength). The initial sliding-window version (pairs sharing episodes) gave flip/sham $= 0.50/0.05$ (3B, scale 0.5), $0.90/0.35$ (scale 1.0), $0.95/0.55$ (scale 4.0) and $0.80/0.55$ at 0.5B (scale 4.0): large perturbations produce nonspecific degradation, which is why gentle scales carry the claim. Candidate answers are scored by first-token log-odds; runs verified free of first-token collisions.

H Multimodal Boundary

A double ablation on 46 landmark-photo episodes, run on two vision-language families: neutral templated contexts remove text leakage (round 1's GPT-written scenes leaked place identity: Qwen2-VL no-image W 0.634); the within-class contrast (true city vs. other cities) removes the city-word class prior.

In Qwen2-VL-2B, image identity is not recoverable from the end-of-encoding word-level workspace in the within-class round (W 0.483 with image vs. 0.475 without; paired $\Delta\mathrm{AUC}$ $+0.008$ $[-0.051, +0.070]$), while ask-time verbal probing reads the image nearly perfectly (0.986 vs. 0.703 without); in the neutral-template round the verbal probe collapses to 0.054 without the image. In LLaVA-1.5-7B the verbal pattern repeats (ask-time V up to 0.968 with

Table 11: Availability-family grids, dynamic members: Rehearsal (emission / decoded reactivation, K=8 rollouts $\times$ 40 tokens) and Leverage (utility-gradient) across models and batteries.

Benchmark	Model	W_{rep}^{emit}	W_{rep}^{dec}	W_J [95% CI]
Evoked	0.5B	0.560	0.518	0.440 [0.373, 0.509]
Evoked	1.5B	0.639	0.684	0.512 [0.443, 0.583]
Evoked	3B	0.586	0.666	0.542 [0.460, 0.624]
Evoked	7B	0.608	0.684	0.506 [0.428, 0.585]
Evoked	olmo7b	0.588	0.652	0.559 [0.480, 0.633]
Decoupled	0.5B	0.449	0.405	0.518 [0.442, 0.595]
Decoupled	1.5B	0.526	0.526	0.505 [0.427, 0.583]
Decoupled	3B	0.485	0.579	0.600 [0.528, 0.674]
Decoupled	7B	0.563	0.578	0.550 [0.470, 0.629]
Decoupled	olmo7b	0.512	0.597	0.532 [0.459, 0.605]
Compositional	0.5B	0.430	0.364	0.489 [0.402, 0.576]
Compositional	1.5B	0.569	0.561	0.573 [0.485, 0.655]
Compositional	3B	0.496	0.532	0.594 [0.508, 0.684]
Compositional	7B	0.565	0.553	0.639 [0.550, 0.725]
Compositional	olmo7b	0.504	0.541	0.515 [0.426, 0.604]

Table 12: Availability-family grids, static Broadcast (ignition) member and its components over the (layer $\times$ position) decodability grid.

Benchmark	Model	W_{ig}	breadth	persistence	sharpness
Evoked	3B	0.639	0.709	0.567	0.531
Evoked	7B	0.569	0.637	0.523	0.487
Decoupled	3B	0.507	0.593	0.463	0.398
Decoupled	7B	0.495	0.557	0.447	0.431
Compositional	3B	0.421	0.492	0.372	0.340
Compositional	7B	0.440	0.486	0.408	0.359

image), but the workspace trace is present: the paired image-minus-ablation contrast is $+0.261$ $[+0.162, +0.359]$ on neutral templates and $+0.131$ $[+0.049, +0.221]$ within-class, with with-image W 0.637 $[0.549, 0.727]$ above chance.

Two conclusions follow. Ask-time verbal access to perceived identity is near-ceiling in both families: perceptually present content behaves like the explicit regime, where the verbal channel is reliable. Whether that content also enters the word-level workspace at encoding is family-dependent, absent in the 2B probe and partial in the 7B LLaVA probe, so workspace-based gating of perceptual content may be viable in some families but cannot be assumed. We therefore scope the paper's core claims to language-derived inference.

I Generator and Family Replications

Table 14 collects the replication rows.

Third generator (gpt-5.6, Decoupled design). The workspace recovery replicates at both scales (W 0.570/0.664) and the verbal failure replicates at 0.5B (V 0.261), while the 7B reporter succeeds on this generator's episodes (V 0.878). On the same Decoupled design, across generators and scales, the verbal channel therefore swings from 0.26 to 0.88 (down to 0.17 including Compositional under the primary generator) while the workspace readout stays within 0.53–0.66; on gpt-4o's weakly evoked bridges both channels attenuate.

Cue density, measured. To turn the sparse/dense description into a variable, an external annotator (gpt-5.6, blind to our hypotheses' direction) counted the distinct cues per episode that would each independently allow the bridge to be inferred, across all three Decoupled-design sets (230 episodes). Cue count predicts 7B reporter success (bridge ranked top-2 by V): pooled Spearman ρ=0.30 (p=3.9×10^{-6}), monotone from 0.32 at $\leq$1 cue to 0.67 at $\geq$4 cues, and the dose-response holds within the primary generator alone (ρ=0.25, p=4×10^{-4}, n=188). A second, independent annotator (gpt-4.1, which generated none of the gpt-5.6 episodes it scores) replicates the dose-response (ρ=0.33

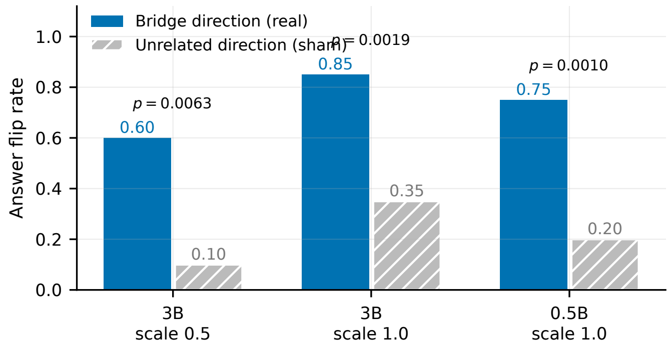

Figure 7: Causal check: steering along the stored bridge's input-side direction (real) flips memory-based answers far more often than an unrelated direction (sham); disjoint episode pairs, exact McNemar p above each pair of bars.

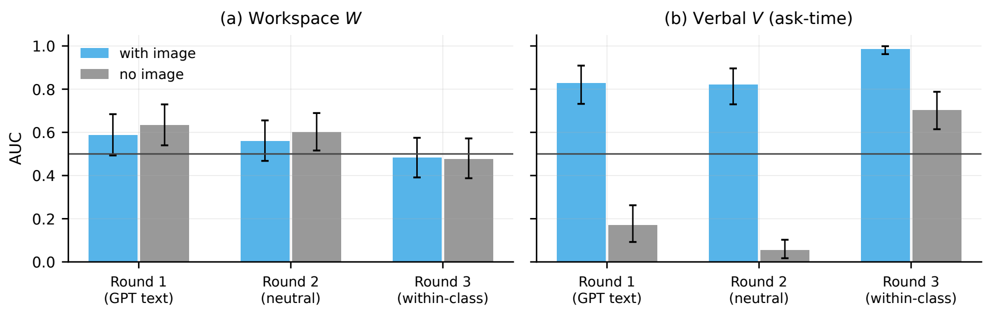

Figure 8: Multimodal double ablation (Qwen2-VL-2B, landmark photos): the end-of-encoding word-level workspace never distinguishes the pictured city within-class (a), while ask-time verbal probing reads the image nearly perfectly and collapses without it (b).

pooled, $p{=}3 \times 10^{-7}$) with rank agreement $\rho{=}0.56$ between annotators. Cue count is a partial account: the gpt-5.6 set's reporter rate (0.905 at mean 3.8 cues) exceeds what count alone predicts relative to the primary generator (0.42 at mean 3.0), so residual generator differences in cue strength remain, and we scope claims accordingly. The workspace readout is insensitive to this variable in the measured range, which is the practical point: the reporter requires cue density it cannot be guaranteed, the readout does not.

J Verbal-Probe Robustness and the Canonical Rating Baseline

Surface-absence confound and within-absent discrimination. On the inferred-regime benchmarks the bridge is nearly the only context-absent candidate ($1{-}\rho$ AUC 0.98–0.99; $72/75$ Evoked episodes have exactly one absent item), so a rule that keeps absent strings solves episode-level selection there without internals. This is a property of the benchmark, not of the channels: in deployment the space of absent concepts is unbounded and bare absence is vacuous. To test what the workspace signal actually carries, we score each episode's bridge against ten foils drawn from other episodes' bridges, filtered to be equally absent from this context. The readout ranks the evoked bridge far above matched absent foils: within-absent AUC 0.763 $[0.717, 0.808]$ on Evoked and 0.689 $[0.625, 0.748]$ on Decoupled at 7B, and 0.697 $[0.643, 0.751]$ on Decoupled even at 0.5B (episode-cluster bootstrap CIs). This rules out bare absence but not, by itself, semantic relatedness: a sentence encoder also separates the bridge from unrelated foils (within-absent $0.914/0.918$). The two explanations come apart on the full candidate sets, where embedding relevance fails on Decoupled ($W{-}E$ $+0.170$ $[+0.072, +0.275]$) while the readout holds, and causally, where patching the readout's directions moves memory-based answers (§5.4). The same test explains the verbal channel's failure mode: the base 0.5B verbal probe separates the bridge from absent foils (0.870) yet ranks it below vivid present distractors in the full candidate set (0.281), i.e. the report mis-prioritizes surface content rather than failing to sense evocation, the machine form of cue-utilization (Koriat, 1997).

Table 13: Multimodal boundary, two vision-language families (46 landmark episodes). Ask-time verbal probing reads the pictured identity nearly perfectly in both models. The word-level workspace trace of the image is family-dependent: absent in Qwen2-VL-2B (within-class contrast $\approx$ no-image ablation) but present in LLaVA-1.5-7B (paired image$-$ablation ΔAUC $+0.261$ $[+0.162, +0.359]$ on neutral templates and $+0.131$ $[+0.049, +0.221]$ within-class).

Model, benchmark variant	Condition	W [95% CI]	V [95% CI]
Qwen2-VL-2B, Round 1 (GPT-written text)	with image	0.586 [0.493, 0.684]	0.827 [0.732, 0.909]
	no image	0.634 [0.540, 0.729]	0.171 [0.092, 0.262]
Qwen2-VL-2B, Round 2 (neutral templates)	with image	0.559 [0.468, 0.655]	0.820 [0.730, 0.896]
	no image	0.602 [0.516, 0.689]	0.054 [0.017, 0.103]
Qwen2-VL-2B, Round 3 (within-class cities)	with image	0.483 [0.391, 0.575]	0.986 [0.962, 0.999]
	no image	0.475 [0.388, 0.572]	0.703 [0.615, 0.788]
LLaVA-1.5-7B, Round 1 (GPT-written text)	with image	0.665 [0.562, 0.765]	0.910 [0.841, 0.966]
	no image	0.400 [0.300, 0.507]	0.417 [0.322, 0.514]
LLaVA-1.5-7B, Round 2 (neutral templates)	with image	0.570 [0.467, 0.671]	0.862 [0.783, 0.930]
	no image	0.309 [0.213, 0.412]	0.220 [0.144, 0.300]
LLaVA-1.5-7B, Round 3 (within-class cities)	with image	0.637 [0.549, 0.727]	0.968 [0.941, 0.987]
	no image	0.506 [0.410, 0.601]	0.702 [0.611, 0.790]

Probe wordings. The importance probe $u(c, x)$ of Eq. 6 (P1) and its paraphrases, verbatim (each followed by "Answer with a single word: yes or no."): P1: "Based only on the passage above, is the concept 'c' one of the most important things to remember in order to answer possible future questions?"; P2: "You can only keep a few notes about this passage. Should 'c' be one of them?"; P3: "Rate the concept 'c': will it be useful later for answering questions about what you just read?". The canonical rating prompt: "On a scale of 1 to 10, where 1 is purely mundane and 10 is extremely important, rate how important it is to remember the concept 'c' in order to answer possible future questions about this passage. Answer with a single number." The rating score is the expected digit under the model's next-token distribution. A length-normalized conditional-likelihood score, $\log p(c \mid x)/|c|$ (one extra forward pass per candidate), is a strong internal baseline: AUC $0.588/0.635/0.635/0.592$ on Explicit/Evoked/Decoupled/Compositional at 7B, comparable to the workspace readout predictively; gated downstream it is likewise statistically indistinguishable from workspace gating on the mixed pool (App. P). On Compositional its cluster CI is $[0.507, 0.677]$ and the paired difference against the static readout is $+0.078$ $[-0.040, +0.188]$ (n=261 items, 4,000-draw episode-cluster bootstrap), so it does not meet the paired recovery criterion that the Leverage readout does ($+0.125$ $[+0.010, +0.244]$, Table 5); head-to-head the two do not separate (W_J-LL $+0.046$ $[-0.056, +0.153]$), so the family-internal claim of §6 is the one the data support. Both internal signals beat the verbal report on inferred content, reinforcing the thesis; the workspace readout is the variant that costs no additional forward passes and carries the causal grounding of §5.4. A deliberative variant ("think briefly about what the passage implies" before a final yes/no) does not rescue the reporter on Decoupled at either scale: the chain-of-thought probe scores AUC 0.453 at 0.5B and 0.493 at 7B (the latter with a 0.91 yes-rate); generation-time deliberation in the probe is not the missing ingredient, consistent with free-running Rehearsal helping only when its rollouts are scored by the readout rather than self-reported (§6.2). Results for the paraphrases and the rating appear in Table 15 and Fig. 9.

The failure of V on the construct-valid benchmark is not an artifact of one prompt wording: all three paraphrases and their ensemble sit at or below chance while every paired $W-V$ contrast excludes zero. The canonical Generative-Agents-style 1–10 rating fails there as well. On the answer-coupled Evoked benchmark at 0.5B, the 1–10 rating is substantially better than the binary probe (0.565), and rating-gated selection matches workspace-gated selection (0.747 vs. 0.760 at k=3, McNemar even). On the construct-valid Decoupled benchmark at 7B the rating probe is at chance predictively (0.493) and rating-gated selection trails workspace-gated selection without reaching significance at n=68 episodes (0.485 vs. 0.529 at k=3); on the 120-episode scaled benchmark it remains at or below chance predictively (0.457 [0.397, 0.519] vs. W 0.599) and trails directionally in gating (0.450 vs. 0.525 at k=3; McNemar $+23/-14$, p=0.19); the significant 7B gating advantage in the main text is over the binary probe. Summary: no verbal variant we tested recovers sparse-cue construct-valid inferred content predictively (the densely cued third-generator setting where the 7B reporter recovers is App. I); gating consequences follow the predictive gap where the task ceiling allows.

Table 14: Generator- and family-independence replications. The $W - V$ dissociation survives an independent benchmark generator (gpt-4o; note the absolute W level is generator-dependent: gpt-4o's bridges are weakly evoked) and a third model family (Mistral-7B). The Explicit rows complete the regime map across families: on stated content the sign flips and the verbal report beats the workspace readout in every family tested. The Decoupled-G3 rows (a third generator, gpt-5.6, whose episodes carry several convergent cues per bridge) show the two channels' stability profiles: the workspace readout stays in a narrow band across all three generators while the verbal report swings from anti-calibrated to well-calibrated with generator and scale.

Setting	W	V	$W - V$ [95% CI]
Evoked-G2 (gpt-4o), Qwen-0.5B-I	0.486	0.282	+0.204 [+0.095, +0.306]*
Evoked-G2 (gpt-4o), Qwen-3B-I	0.488	0.223	+0.265 [+0.160, +0.365]*
Evoked-G2 (gpt-4o), Qwen-7B-I	0.490	0.381	+0.109 [-0.004, +0.215]
Evoked-G2 (gpt-4o), OLMo-1B (RLVR)	0.512	0.382	+0.130 [+0.026, +0.238]*
Decoupled, Mistral-7B-I	0.672	0.481	+0.191 [+0.089, +0.299]*
Compositional, Mistral-7B-I	0.596	0.322	+0.274 [+0.172, +0.372]*
Explicit, Mistral-7B-I	0.532	0.622	-0.089 [-0.175, -0.009]*
Explicit, OLMo-2-7B-I	0.492	0.622	-0.129 [-0.209, -0.052]*
Decoupled-G3 (gpt-5.6), Qwen-0.5B-I	0.570	0.261	+0.309 [+0.198, +0.419]*
Decoupled-G3 (gpt-5.6), Qwen-7B-I	0.664	0.878	-0.213 [-0.311, -0.116]*

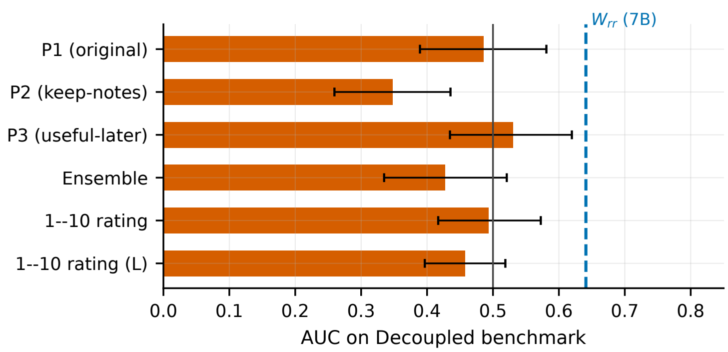

Figure 9: No verbal variant recovers sparse-cue construct-valid inferred content: three importance-probe paraphrases, their ensemble, and the canonical 1–10 rating (both Decoupled sizes) all sit at or below chance, against the workspace reference (dashed).

K Surface-Feature Fusion Baseline

The fusion proxy (rank-average of recency, frequency, and embedding relevance) underperforms workspace gating in both settings; on evoked-bridge benchmarks this is expected, since the decisive item never occurs in the text and has no surface features to fuse.

L Metacognitive Alignment: Details

Training set: 1,040 importance-probe examples (top-2 workspace-ranked items per episode labeled *yes*) from Explicit, Evoked, and the gpt-4o benchmark, measured on OLMo-2-1B-Instruct itself; AdamW, lr 10^{-5}, batch 4, 2 epochs (520 optimizer steps), bf16, gradient checkpointing, single A5000, $\sim$20 minutes. No-harm checks on held-out Decoupled: full-context QA $0.426 \rightarrow 0.397$ (exact McNemar $+0/-2$, p=0.50), no-memory $0.250 \rightarrow 0.250$. The post-alignment reporter also transfers its calibration to Compositional ($0.267 \rightarrow 0.580$), exceeding the static readout there (0.491).

Second-family replication (Qwen2.5-0.5B-Instruct). The identical recipe (same benchmarks, labeling rule, and hyperparameters; 520 steps; labels from Qwen-0.5B's own workspace ranking) gives held-out Decoupled V: $0.281 \rightarrow 0.668$ with W unchanged ($0.526 \rightarrow 0.524$), and Compositional V: $0.172 \rightarrow 0.451$ with W unchanged ($0.416 \rightarrow 0.409$). Full-context QA on Decoupled at the matched 24-token budget moves from 0.353 to 0.309 (exact McNemar $+5/-8$, p=0.58; no-memory floor 0.147 unchanged). Note the Compositional endpoint: at 0.5B the teacher W is itself at chance on this benchmark, and the repaired reporter plateaus at the teacher's level rather than above it, as self-distillation predicts.

Table 15: Verbal-probe paraphrase robustness and the canonical 1–10 rating baseline (Qwen; W reference at 7B on the Decoupled benchmark: 0.641). Rating-gated selection runs use their own generation budgets, so compare within-run columns only.

Importance probe	V AUC [95% CI]	paired $W - V$ [95% CI]
P1 (original)	0.486 [0.389, 0.581]	+0.155 [+0.053, +0.263]
P2 (keep-notes)	0.347 [0.259, 0.436]	+0.294 [+0.193, +0.394]
P3 (useful-later)	0.530 [0.435, 0.620]	+0.111 [+0.009, +0.215]
3-prompt ensemble	0.428 [0.335, 0.521]	+0.214 [+0.113, +0.318]
1–10 rating, Decoupled 7B	0.493 [0.417, 0.573]	—
1–10 rating, Decoupled-L 7B	0.457 [0.397, 0.519]	—
1–10 rating, Evoked 0.5B	0.565 [0.493, 0.638]	—

Table 16: Surface-feature fusion baseline (rank-average of recency, frequency, and embedding relevance, an admission-control proxy without internals; generation budgets as marked per row, so compare within-run columns).

Setting	Policy	k=2	k=3
Qwen-0.5B, Evoked (24 tok)	workspace	0.600	0.760
Qwen-0.5B, Evoked (24 tok)	verbal	0.413	0.480
Qwen-0.5B, Evoked (24 tok)	fusion	0.213	0.293
Qwen-7B, Decoupled (64 tok)	workspace	0.338	0.529
Qwen-7B, Decoupled (64 tok)	verbal	0.353	0.426
Qwen-7B, Decoupled (64 tok)	fusion	0.235	0.382

Label noise and the teacher ceiling. The teacher's top-2 labels are only 0.21–0.30 precise on the training benchmarks, so the student learns from a noisy but directionally informative signal; that students can end above the teacher's own held-out ranking is the familiar consequence of averaging over label noise, not a contradiction of the distillation account.

General-capability check. Because the fine-tune touches the importance-report format, we verify it does not degrade general abilities: Table 17 evaluates every checkpoint on MMLU (full test set, 14,042 items, letter log-likelihood), ARC-Easy/Challenge (length-normalized log-likelihood), and GSM8K (5-shot greedy, fixed 500-item subsample), with identical prompts and item order across checkpoints so the before/after columns are paired. Differences are within ± 0.01 on MMLU and GSM8K in both families and no benchmark degrades beyond noise (ARC moves are $+0.01$ to $+0.03$ in the tuned direction at 0.5B); $\sim$500 narrow self-distillation steps repair the reporter without touching general capability, consistent with the unchanged workspace channel and QA no-harm checks above.

Table 17: Metacognitive alignment does not degrade general capability. Identical prompts and item order across checkpoints; MMLU is the full test set; GSM8K is a fixed 500-item subsample (5-shot).

Checkpoint	ARC-E	ARC-C	MMLU	GSM8K
Qwen2.5-0.5B-I (base)	0.593	0.329	0.460	0.308
+ metacog (seed 0)	0.622	0.343	0.459	0.308
+ metacog (seed 1)	0.615	0.337	0.460	0.300
+ metacog (seed 2)	0.615	0.338	0.461	0.308
OLMo-2-1B-I (base)	0.705	0.431	0.410	0.610
+ metacog (seed 1)	0.708	0.427	0.401	0.620
+ metacog (seed 2)	0.707	0.423	0.398	0.622

Seed variance. Across three fine-tuning seeds per family, held-out Decoupled calibration reaches 0.654 on average for Qwen-0.5B (0.540–0.754) and 0.617 for OLMo-1B (0.523–0.671); every seed ends above the 0.28–0.33 baselines, with one partial-repair seed in each family, so the effect is robust in direction and variable in magnitude.

Controls. Three checks rule out the trivial explanations of the gain. (i) *Label shuffle*: fine-tuning with identical prompts, steps, and yes/no frequencies but permuted labels leaves held-out Decoupled V at chance in both families (Qwen-0.5B 0.416 vs. 0.668 real; OLMo-1B 0.382 vs. 0.671 real): de-saturating the yes-heavy probe format does not by itself produce calibration. (ii) *Label source*: distilling from embedding-relevance-derived labels instead of workspace-derived ones, same recipe, fails on held-out Decoupled (V 0.458): the workspace signal specifically, not generic salience, carries the repair. (iii) *Absence shortcut*: the aligned reporter does not simply answer yes to context-absent strings; against ten equally absent foils per episode it ranks the true bridge at within-absent AUC 0.900 $[0.865, 0.932]$.

M Reproducibility

Released artifact files carry internal identifiers in their filenames: `v1` (Explicit), `v2` (Evoked), `v3` (Compositional), `v4` (Decoupled), and `v4xl` (Decoupled-L). All experiments ran on $8\times$ RTX A5000 (24 GB) with PyTorch 2.13/bf16; greedy decoding everywhere except Rehearsal rollouts (seeded per episode). Case-variant token candidates are the first tokens of the leading-space, capitalized, and bare variants of each concept string, applied uniformly across items. The provenance test ρ (Eq. 4) is implemented as a case-insensitive substring match of the concept string against the source context. The yes/no token sets Y, N of Eq. 6 are the first tokens of {yes, Yes, YES} and {no, No, NO} with and without leading space. The verbal probe and its paraphrases are listed in App. J; the grader accepts a whole-word match of the gold answer, and strict re-grading changes no policy cell by more than 0.03 and never reorders policies. Benchmarks, per-item measurements, per-episode downstream records, and all analysis scripts are released with the supplementary material (public repository upon publication); every table and figure in this paper is regenerated from the raw measurement records by a single released script.

N Naturalistic-Stream Boundary: LoCoMo and LongMemEval

The benchmarks of §3.2 are synthetic by design. We evaluated the same write-admission policies on the two standard naturalistic long-term-memory benchmarks, LoCoMo (Maharana et al., 2024) and LongMemEval (Wu et al., 2025), and report the result as a boundary: these streams do not contain the construct this paper measures, and on them no question-blind content channel, including the verbal importance probe (Eq. 6), separates from random selection.

Setup. LoCoMo: each conversation session is a context x, the dataset's own per-session observation annotations are the candidate notes a host memory system would propose, and single-hop questions whose annotated evidence lies in one session are probes (90 episodes, 120 questions, 829 notes, budget k=3 of $\sim$9 notes). LongMemEval (oracle variant): the evidence session of each single-session question is the context and its turns are the candidates (118 episodes/questions, budget k=3 of $\sim$12 turns; the turn-level evidence flags give the oracle). Sentence-level candidates are scored by lifting the single-concept readout: W of a note is the maximum over its content words (*max*), over its context-absent words only (*absent*; the construct-faithful choice, since those words are the note's paraphrased part), or the mean over words (*mean*, used for verbatim turns). Verbal and embedding scores apply Eq. 6 and E to the note text.

Grading. Because gold answers are free-form, token-F1 misgrades paraphrased answers; an LLM judge (gpt-4.1-mini, temperature 0, yes/no entailment of the gold answer) disagrees with F1 on 22–30% of responses. All conclusions below use judge grading; the only policy contrast that was significant under F1 (verbal over max-lifted workspace at 0.5B, p=0.039) disappears under the judge (p=0.87), i.e. it was a grading artifact. A second, independent judge (gpt-5.6) is systematically stricter, tracking token-F1's absolute levels (raw agreement with the first judge 74%), and preserves every relative conclusion: under all three graders no question-blind pair separates (minimum pairwise p=0.053 lenient, 0.064 strict), so the null is grading-robust.

Lifting check. The naive max lifting is itself unsound: reciprocal ranks are discrete, 43–53% of episodes tie across the top-k boundary (selection then follows annotation order), and at 0.5B its selection-level evidence containment falls below random. The absent-words lifting removes the artifact; all liftings agree on the conclusions below.

Results. Table 18 reports accuracy under judge grading.

Why: the construct is structurally absent. LongMemEval turn candidates are verbatim substrings of the context, so every candidate has ρ=1 by construction; LoCoMo notes are annotator summaries of stated dialogue content (median 83% of a note's content words appear verbatim in the session). Neither stream contains a silently inferred, load-bearing candidate in the sense of Eq. 4, so the regime that separates the paper's channels cannot occur, and a well-behaved workspace readout is expected to go null here, as it does. Three take-aways: (i) the provenance-controlled benchmarks of §3.2 are not a convenience but the instrument required to measure the construct, and we release them as such; (ii) on surface-only naturalistic streams the practical bottleneck is question-blind admission itself (oracle gap), not the choice of scoring channel; (iii) a naturalistic benchmark whose questions hinge on silent inferences,

Table 18: Write-admission on naturalistic streams (LLM-judge grading, k=3). LoCoMo uses the absent-words lifting. No pairwise McNemar between question-blind policies reaches significance in any column (minimum p=0.053 over all 40 tests, for workspace over recency on LongMemEval at 0.5B); the evidence oracle roughly doubles every policy.

	LoCoMo		LongMemEval	
Policy	0.5B	7B	0.5B	7B
Workspace	0.292	0.317	0.271	0.331
Verbal	0.300	0.300	0.220	0.305
Embedding	0.317	0.258	0.254	0.322
Recency	0.350	0.383	0.186	0.288
Random	0.308	0.367	0.263	0.381
Evidence oracle	0.583	0.658	0.602	0.754
No memory	0.117	0.225	0.085	0.136
Full context	0.767	0.875	0.686	0.983

which existing suites do not provide (LoCoMo's multi-hop questions almost always span sessions; only $12/282$ have single-session evidence), is the missing instrument for testing inference-gated memory in the wild.

Naturalistic carriers: the dissociation is content, not style. As a positive control for that missing instrument, we embed each of the 68 Decoupled episodes into a real LoCoMo session: the episode's paragraph is attributed to one speaker and appended as the final turn of a $\geq$10-turn session (mean 583 words of context). Candidates, labels, and probes are unchanged; only the carrier changes from synthetic prose to real dialogue. At 7B the workspace readout is essentially unattenuated (W: 0.631 $[0.559, 0.700]$ wrapped vs. 0.641 original) while the verbal report remains at or below chance (V: 0.425 wrapped vs. 0.486 original), and the paired dissociation on wrapped episodes is significant ($W - V$ $+0.206$ $[+0.101, +0.309]$). At 0.5B, where the original Decoupled signal is already marginal (0.526), wrapping dilutes it to chance (0.502). Planting the paragraph mid-session instead of last attenuates but preserves the 7B readout (W 0.583 $[0.506, 0.658]$; V 0.343), so the effect is not an artifact of end-of-context recency. The dissociation is therefore carried by the provenance of the content, not by the style of the surrounding text: when silently inferred, load-bearing content occurs inside real dialogue, the workspace advantage reappears at capable scale.

O Pre-Specified Predictions and Deviations

The four primary contrasts specified in the project protocol before the family experiments are $W - V$ on the construct-valid Decoupled design at each size, with Bonferroni-corrected 98.75% episode-cluster CIs. At 0.5B the contrast is $+0.244$ (CI $[+0.130, +0.355]$), at 3B $+0.340$ (CI $[+0.222, +0.451]$), and at 7B $+0.155$ (CI $[+0.028, +0.286]$), each excluding zero after correction; at 1.5B it is $+0.115$ (CI $[-0.019, +0.258]$) on the original 68-episode set and $+0.113$ (CI $[+0.012, +0.213]$, excluding zero) on the 120-episode scaled set, which was constructed for the knife-edge 1.5B case before these contrasts were computed; we report both intervals so the reader can apply either convention. Pre-specified before the family experiments: (P1) the Leverage readout exceeds the static readout on compositional bridges: **held** at Qwen 7B (paired $+0.125$, CI excludes 0), directional but not significant at Qwen 3B ($+0.096$, CI includes 0), **failed** on OLMo-2-7B (0.515); (P2) dynamic members do not regress on evoked bridges: **held** for Rehearsal (improves), **failed** for Leverage (0.506 on Evoked); we report the family as complementary rather than uniformly dominant. Other deviations: the 7B verbal "catch-up" observed on the earlier answer-coupled Evoked pilot benchmark disappeared on construct-valid benchmarks (it rode on answer coupling and a saturated yes-regime: 122/165 items with P(yes)$>$0.99); the initial sliding-window causal pairing was replaced by disjoint pairs after audit; the 0.5B/1.5B downstream cells on Decoupled-class tasks are composition-limited and were not part of any claim.

P Provenance-Routed Gating: Full Results and Boundaries

Full mixed-pool comparisons. All numbers in Table 3 and below come from a single run generation (one device, per-benchmark budgets fixed; earlier-generation records are retained in the artifact). Pairwise exact McNemar tests (discordant counts $+a/-b$ favor the first policy; the family below comprises 24 tests, and under Holm correction the 0.5B k=3 workspace-vs-probe contrast remains significant while the 7B k=3 contrast is marginal, so we treat the former as confirmatory and the rest as consistent evidence). Workspace vs. binary probe: 0.5B k=2 $+42/-30$

(p=0.19), k=3 $+47/-18$ (p=0.0004); 7B k=2 $+47/-36$ (p=0.27), k=3 $+45/-22$ (p=0.0067). Workspace vs. embedding: n.s. in all four cells (best 7B k=3 $+42/-27$, p=0.09). Workspace vs. 1–10 rating: n.s. in all four cells, workspace ahead in three (7B k=3 $+44/-34$, p=0.31). Workspace vs. conditional likelihood (post-hoc baseline added in revision, not part of the pre-specified family): n.s. in all four cells (0.5B $+20/-32$ p=0.13, $+21/-31$ p=0.21; 7B $+32/-42$ p=0.30, $+29/-24$ p=0.58); the likelihood gate itself beats the binary probe (0.5B $+48/-24$ p=0.0063, $+56/-17$ $p<10^{-4}$; 7B $+56/-35$ p=0.035, $+47/-29$ p=0.051) and on the construct-valid Decoupled-L cells sits at 0.442/0.500 against workspace 0.425/0.525 (n.s.), with workspace, not the likelihood gate, matching the oracle at k=3. Routed vs. binary probe: 0.5B k=3 $+49/-24$ (p=0.0046). Routed vs. workspace: best at the tight 7B budget ($+44/-41$, n.s.) and worse at the loose one ($+25/-45$, p=0.023), which is why the single-channel gate is the default. Workspace and routed beat recency in every cell ($p<10^{-4}$). Cross-scale pooled contrasts are exploratory (episodes repeat across scales) and consistent with the per-scale tests above.

The absence heuristic: a three-act diagnostic. Because each episode plants exactly one inference, bare absence is nearly an oracle proxy on the released benchmarks ($1-\rho$ AUC 0.98–0.99); the policy "keep context-absent items first" is PRG's router run without a discriminator. Table 19 follows it through three candidate-set regimes on one base (n=68 Decoupled episodes, k=3, 64-token budget): on the single-planted set it is merely competitive; its real advantage is on the mixed pool at tight budgets (0.629 vs. 0.502 at k=2), where construction guarantees the absent item is the target. With unrelated absent foils it collapses and surface semantics (embedding) suffice; with same-category confusable foils, the conservative hard case, embedding saturates and the evocation-sensitive channels lead (directional at this n). Notably, the verbal probe also works as a tie-break once provenance has removed the surface competitors (0.471), consistent with its failure mode being mis-prioritization against vivid present content rather than insensitivity to evocation (App. J); the workspace readout achieves the same without probe passes. The composite is the two-stage design of §4: provenance narrows the pool; among confusable absent candidates no single tie-break dominates at these n's, and the readout is the channel that adds no probe cost, with its significant advantage appearing in the full-pool ranking (see the powered replication below).

Table 19: Absence-heuristic diagnostic across candidate-set regimes: the same 68 Decoupled episodes at 7B, k=3, 64-token budget in every column (oracle identical across columns by construction). "absent+X" = keep context-absent candidates first, tie-break by X. On the mixed pool at the tight budget the heuristic's advantage is larger (0.629 vs. workspace 0.502 at k=2), which is the construction artifact discussed in the text. A pooled 188-episode replication of the confusable regime is analyzed in the text.

Policy	single-planted	+4 unrelated foils	+4 confusable foils
absent+embedding	0.500	0.162	0.441
absent+workspace	0.515	0.162	0.456
absent+verbal	—	—	0.471
Embedding	0.471	**0.485**	0.426
Workspace	**0.529**	0.353	**0.471**
Oracle	0.485	0.485	0.485

Powered replication of the confusable regime. The 68-episode confusable set above generated the hypothesis; to test it we built a fresh 120-episode battery by the same recipe (four same-category absent foils per episode) and pre-designated two endpoints: the full-pool selection-signal contrast (workspace vs. embedding paired AUC, directional) and the gated-QA McNemar grid. The selection-signal endpoint replicates on the fresh set alone: paired ΔAUC $+0.064$, one-sided p=0.026 (two-sided CI $[-0.001, +0.126]$; 1,052 items), giving a pooled estimate over all 188 episodes of workspace 0.635 $[0.596, 0.675]$ vs. embedding 0.577 $[0.545, 0.611]$, paired $+0.058$ $[+0.007, +0.108]$ (episode-cluster bootstrap over 1,659 items). Restricted to the absent candidates alone both signals discriminate (workspace 0.654 $[0.611, 0.698]$; embedding 0.704 $[0.675, 0.735]$): the foils are semantically related by construction, so surface similarity is informative once provenance is fixed, and what the embedder cannot do is the full-pool ranking, where it mis-prioritizes vivid present content. The behavioral endpoint is null at this power: converting two- or three-item selections into 64-token answers adds noise that dominates, and no pairwise policy contrast in the pooled grid reaches significance. Gated QA at k=3 puts workspace first among content policies (0.495; largest contrast, over embedding, $+29/-16$, p=0.073) with no detectable difference from the true-label oracle (0.495 vs. 0.511, $+26/-29$). On the routed slice itself no tie-break separates from any other (absent+workspace vs. absent+embedding $+18/-15$ at k=2 and $+11/-11$ at k=3; absent+verbal vs. absent+embedding p=0.064 and p=0.63), so at these n's the within-absent discriminators are statistically interchangeable and the readout is the one that costs no probe passes. The largest nominal cell, absent→verbal ahead of workspace at k=2 ($+9/-21$, p=0.043 uncorrected, one of twelve comparisons, n=120), receives the same skepticism we apply to every uncorrected cell in this grid; we note its direction is the mechanism's prediction (once provenance routing removes the vivid present competitors, the verbal report recovers) and that the k=2 truncation is noise-dominated (the true-label oracle itself falls to 0.394 there, below several heuristics).

Canonical 1–10 rating. In the single-generation runs the rating reaches 0.343/0.410 (0.5B) and 0.491/0.601 (7B), never separating from workspace gating (which is numerically ahead in three of four cells) and costing one probe pass per candidate where the readout costs none; on the construct-valid slice it is predictively at or below chance (App. J). A rating-routed PRG variant (ρ=1 route scored by the rating) does not improve selection and we do not pursue it.

Calibration ablation. Under z-normalization the two routes are not comparable: reciprocal ranks are heavy-tailed near zero while cosine similarities are approximately Gaussian, so inferred items receive systematically negative z-scores and lose admission slots. Load-bearing containment at k=2 (fraction of episodes whose kept set includes the load-bearing item, offline over the pool): workspace-only 0.488/0.555 (0.5B/7B), z-routed 0.456/0.435, rank-routed 0.583/0.611. Rank calibration recovers the routing advantage that z-scoring destroys; the corresponding end-to-end accuracies appear in Table 3.

Calibration-stream robustness. Three checks on the containment metric (load-bearing containment at k=2, mixed pool). (i) *Window size*: with calibration populations of N=10/25/50/100/200 items per route (200 resamples each), mean containment is flat at 0.56–0.57 (0.5B) and 0.60–0.61 (7B) against full-pool values of 0.583/0.611; only the resample variance shrinks with N (95% spread $[0.38, 0.79]$ at N=10 narrowing to $[0.51, 0.61]$ at N=200, 0.5B). A running window of ∼50–100 past writes therefore suffices in deployment. (ii) *Held-out calibration*: fitting the CDFs on a random episode half and evaluating on the other half gives 0.575/0.613, matching the same-pool values, so the pooled analysis carries no transductive advantage. (iii) *Distribution shift*: CDFs fitted on Explicit ∪ Evoked and applied unchanged to Decoupled-L give 0.458/0.550 on that subset, above the 0.400/0.475 obtained by calibrating on Decoupled-L alone; a broader calibration mix helps rather than hurts.

Harvest boundary. The generative extension of the gate (§4) decodes candidate concepts directly from the workspace instead of scoring writer-proposed ones. With contrastive denoising against a neutral context and whole-word filtering, recovery of the true silent bridge among the top-H decoded concepts at 7B is 0.027/0.040/0.080/0.107 at H=8/25/50/100 on Evoked and 0.029/0.029/0.059/0.103 on Decoupled: far below the discriminative readout's utility. The plain logit lens supports discriminative use (scoring a given candidate) but not generative use (surfacing the candidate from a ∼150k-token vocabulary, where the bridge's median decode rank is ∼290); closing this gap is an open problem for stronger readouts.

Q Cognitive-Science Correspondences

The dissociation maps onto the metamemory literature: judgments of learning ride on surface cues (fluency, vividness) and fail for inference-demanding material (Koriat, 1997; Nelson and Narens, 1990); our V nominating vivid distractors is the machine form of that cue-utilization bias (cf. the von Restorff isolation effect). Information that is present, causally used, and unreportable is the signature of blindsight (Weiskrantz, 1986); a reporting module fluently confabulating reasons it cannot access recalls the split-brain interpreter (Gazzaniga, 2000). M_s imports the meta-d′/d′ logic of Fleming and Lau (2014): humans average $M \approx 0.8$; our models are bimodal (M>1 explicit, M_s<0 inferential). The family members correspond to attentional selection (Spotlight), global-workspace ignition and broadcast (Broadcast; Dehaene et al., 2011), with the twist that in these models global decodability does not entail reportability. Hippocampal replay (Rehearsal; Wilson and McNaughton, 1994), and eligibility-trace-like credit for future outcomes (Leverage; cf. synaptic tagging and capture, Frey and Morris, 1997). Each analogy's tightness and known disanalogies are catalogued in the released supplementary notes; we use them as organizing hypotheses, not claims of mechanistic identity.